

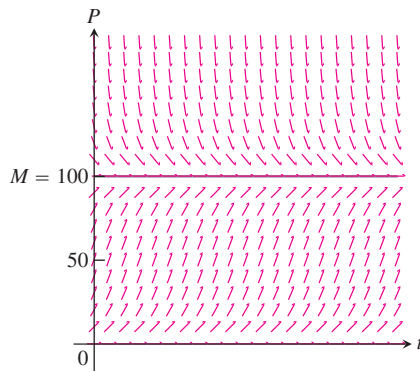


FIGURE 9.25 A slope field for the logistic differential equation $dP/dt = 0.001(100 - P)P$ (Example 2).

- Draw and describe a slope field for the differential equation.
- Use Euler's method with step size $dt = 1$ to estimate the population size in 20 years.
- Find a logistic growth analytic solution $P(t)$ for the population and draw its graph.
- When will the bear population reach 50?

Solution

- Slope field.* The carrying capacity is 100, so $M = 100$. The solution we seek is a solution to the following differential equation.

$$\frac{dP}{dt} = 0.001(100 - P)P$$

Figure 9.25 shows a slope field for this differential equation. There appears to be a horizontal asymptote at $P = 100$. The solution curves fall toward this level from above and rise toward it from below.

- Euler's method.* With step size $dt = 1$, $t_0 = 0$, $P(0) = 10$, and

$$\frac{dP}{dt} = f(t, P) = 0.001(100 - P)P,$$

we obtain the approximations in Table 9.6, using the iteration formula

$$P_n = P_{n-1} + 0.001(100 - P_{n-1})P_{n-1}.$$

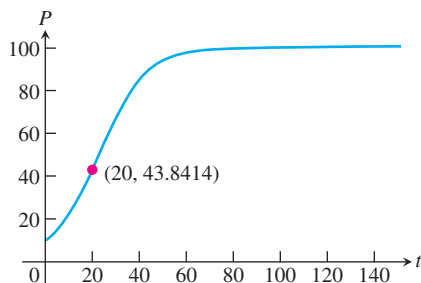


FIGURE 9.26 Euler approximations of the solution to $dP/dt = 0.001(100 - P)P$, $P(0) = 10$, step size $dt = 1$.

TABLE 9.6 Euler solution of $dP/dt = 0.001(100 - P)P$, $P(0) = 10$, step size $dt = 1$

t	P (Euler)	t	P (Euler)
0	10		
1	10.9	11	24.3629
2	11.8712	12	26.2056
3	12.9174	13	28.1395
4	14.0423	14	30.1616
5	15.2493	15	32.2680
6	16.5417	16	34.4536
7	17.9222	17	36.7119
8	19.3933	18	39.0353
9	20.9565	19	41.4151
10	22.6130	20	43.8414

There are approximately 44 grizzly bears after 20 years. Figure 9.26 shows a graph of the Euler approximation over the interval $0 \leq t \leq 150$ with step size $dt = 1$. It looks like the lower curves we sketched in Figure 9.24.

- (c) *Analytic solution.* We can assume that $t = 0$ when the bear population is 10, so $P(0) = 10$. The logistic growth model we seek is the solution to the following initial value problem.

$$\text{Differential equation: } \frac{dP}{dt} = 0.001(100 - P)P$$

$$\text{Initial condition: } P(0) = 10$$

To prepare for integration, we rewrite the differential equation in the form

$$\frac{1}{P(100 - P)} \frac{dP}{dt} = 0.001.$$

Using partial fraction decomposition on the left-hand side and multiplying both sides by 100, we get

$$\left(\frac{1}{P} + \frac{1}{100 - P} \right) \frac{dP}{dt} = 0.1$$

$$\ln |P| - \ln |100 - P| = 0.1t + C \quad \text{Integrate with respect to } t.$$

$$\ln \left| \frac{P}{100 - P} \right| = 0.1t + C$$

$$\ln \left| \frac{100 - P}{P} \right| = -0.1t - C \quad \ln \frac{a}{b} = -\ln \frac{b}{a}$$

$$\left| \frac{100 - P}{P} \right| = e^{-0.1t - C} \quad \text{Exponentiate.}$$

$$\frac{100 - P}{P} = (\pm e^{-C})e^{-0.1t}$$

$$\frac{100}{P} - 1 = Ae^{-0.1t} \quad \text{Let } A = \pm e^{-C}.$$

$$P = \frac{100}{1 + Ae^{-0.1t}}. \quad \text{Solve for } P.$$

This is the general solution to the differential equation. When $t = 0$, $P = 10$, and we obtain

$$10 = \frac{100}{1 + Ae^0}$$

$$1 + A = 10$$

$$A = 9.$$

Thus, the logistic growth model is

$$P = \frac{100}{1 + 9e^{-0.1t}}.$$

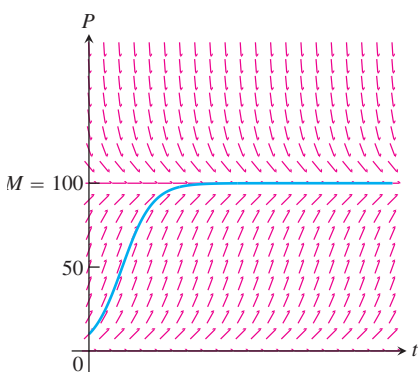


FIGURE 9.27 The graph of

$$P = \frac{100}{1 + 9e^{-0.1t}}$$

superimposed on the slope field in Figure 9.25 (Example 2).

Its graph (Figure 9.27) is superimposed on the slope field from Figure 9.25.

(d) When will the bear population reach 50? For this model,

$$\begin{aligned}
 50 &= \frac{100}{1 + 9e^{-0.1t}} \\
 1 + 9e^{-0.1t} &= 2 \\
 e^{-0.1t} &= \frac{1}{9} \\
 e^{0.1t} &= 9 \\
 t &= \frac{\ln 9}{0.1} \approx 22 \text{ years.}
 \end{aligned}$$

The solution of the general logistic differential equation

$$\frac{dP}{dt} = r(M - P)P$$

can be obtained as in Example 2. In Exercise 10, we ask you to show that the solution is

$$P = \frac{M}{1 + Ae^{-rMt}}.$$

The value of A is determined by an appropriate initial condition.

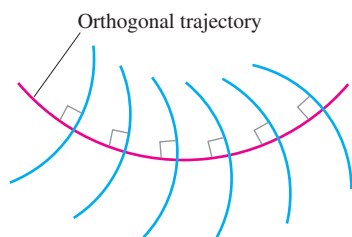


FIGURE 9.28 An orthogonal trajectory intersects the family of curves at right angles, or orthogonally.

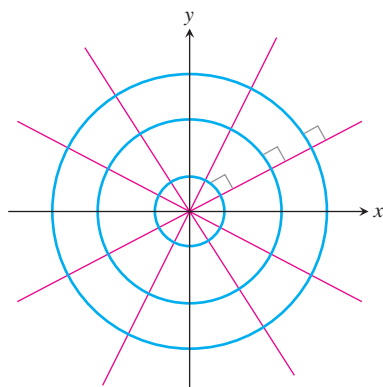


FIGURE 9.29 Every straight line through the origin is orthogonal to the family of circles centered at the origin.

Orthogonal Trajectories

An **orthogonal trajectory** of a family of curves is a curve that intersects each curve of the family at right angles, or *orthogonally* (Figure 9.28). For instance, each straight line through the origin is an orthogonal trajectory of the family of circles $x^2 + y^2 = a^2$, centered at the origin (Figure 9.29). Such mutually orthogonal systems of curves are of particular importance in physical problems related to electrical potential, where the curves in one family correspond to flow of electric current and those in the other family correspond to curves of constant potential. They also occur in hydrodynamics and heat-flow problems.

EXAMPLE 3 Finding Orthogonal Trajectories

Find the orthogonal trajectories of the family of curves $xy = a$, where $a \neq 0$ is an arbitrary constant.

Solution The curves $xy = a$ form a family of hyperbolas with asymptotes $y = \pm x$. First we find the slopes of each curve in this family, or their dy/dx values. Differentiating $xy = a$ implicitly gives

$$x \frac{dy}{dx} + y = 0 \quad \text{or} \quad \frac{dy}{dx} = -\frac{y}{x}.$$

Thus the slope of the tangent line at any point (x, y) on one of the hyperbolas $xy = a$ is $y' = -y/x$. On an orthogonal trajectory the slope of the tangent line at this same point must be the negative reciprocal, or x/y . Therefore, the orthogonal trajectories must satisfy the differential equation

$$\frac{dy}{dx} = \frac{x}{y}.$$

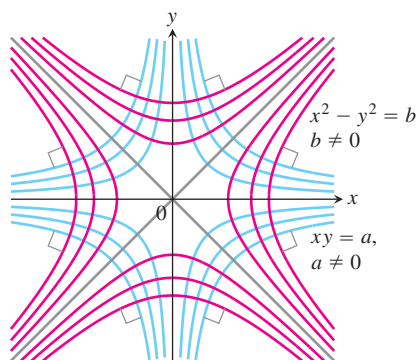


FIGURE 9.30 Each curve is orthogonal to every curve it meets in the other family (Example 3).

This differential equation is separable and we solve it as in Section 9.1:

$$\begin{aligned}
 y \, dy &= x \, dx && \text{Separate variables.} \\
 \int y \, dy &= \int x \, dx && \text{Integrate both sides.} \\
 \frac{1}{2}y^2 &= \frac{1}{2}x^2 + C \\
 y^2 - x^2 &= b, && (6)
 \end{aligned}$$

where $b = 2C$ is an arbitrary constant. The orthogonal trajectories are the family of hyperbolas given by Equation (6) and sketched in Figure 9.30. ■

EXERCISES 9.5

- Coasting bicycle** A 66-kg cyclist on a 7-kg bicycle starts coasting on level ground at 9 m/sec. The k in Equation (1) is about 3.9 kg/sec.
 - About how far will the cyclist coast before reaching a complete stop?
 - How long will it take the cyclist's speed to drop to 1 m/sec?
- Coasting battleship** Suppose that an Iowa class battleship has mass around 51,000 metric tons (51,000,000 kg) and a k value in Equation (1) of about 59,000 kg/sec. Assume that the ship loses power when it is moving at a speed of 9 m/sec.
 - About how far will the ship coast before it is dead in the water?
 - About how long will it take the ship's speed to drop to 1 m/sec?
- The data in Table 9.7 were collected with a motion detector and a CBL™ by Valerie Sharritts, a mathematics teacher at St. Francis DeSales High School in Columbus, Ohio. The table shows the distance s (meters) coasted on in-line skates in t sec by her daughter Ashley when she was 10 years old. Find a model for Ashley's position given by the data in Table 9.7 in the form of Equation (2). Her initial velocity was $v_0 = 2.75$ m/sec, her mass $m = 39.92$ kg (she weighed 88 lb), and her total coasting distance was 4.91 m.
- Coasting to a stop** Table 9.8 shows the distance s (meters) coasted on in-line skates in terms of time t (seconds) by Kelly Schmitzer. Find a model for her position in the form of Equation (2). Her initial velocity was $v_0 = 0.80$ m/sec, her mass $m = 49.90$ kg (110 lb), and her total coasting distance was 1.32 m.
- Guppy population** A 2000-gal tank can support no more than 150 guppies. Six guppies are introduced into the tank. Assume that the rate of growth of the population is

$$\frac{dP}{dt} = 0.0015(150 - P)P,$$

where time t is in weeks.

TABLE 9.7 Ashley Sharritts skating data

t (sec)	s (m)	t (sec)	s (m)	t (sec)	s (m)
0	0	2.24	3.05	4.48	4.77
0.16	0.31	2.40	3.22	4.64	4.82
0.32	0.57	2.56	3.38	4.80	4.84
0.48	0.80	2.72	3.52	4.96	4.86
0.64	1.05	2.88	3.67	5.12	4.88
0.80	1.28	3.04	3.82	5.28	4.89
0.96	1.50	3.20	3.96	5.44	4.90
1.12	1.72	3.36	4.08	5.60	4.90
1.28	1.93	3.52	4.18	5.76	4.91
1.44	2.09	3.68	4.31	5.92	4.90
1.60	2.30	3.84	4.41	6.08	4.91
1.76	2.53	4.00	4.52	6.24	4.90
1.92	2.73	4.16	4.63	6.40	4.91
2.08	2.89	4.32	4.69	6.56	4.91

- Find a formula for the guppy population in terms of t .
 - How long will it take for the guppy population to be 100? 125?
- Gorilla population** A certain wild animal preserve can support no more than 250 lowland gorillas. Twenty-eight gorillas were known to be in the preserve in 1970. Assume that the rate of growth of the population is

$$\frac{dP}{dt} = 0.0004(250 - P)P,$$

where time t is in years.

TABLE 9.8 Kelly Schmitzer skating data

t (sec)	s (m)	t (sec)	s (m)	t (sec)	s (m)
0	0	1.5	0.89	3.1	1.30
0.1	0.07	1.7	0.97	3.3	1.31
0.3	0.22	1.9	1.05	3.5	1.32
0.5	0.36	2.1	1.11	3.7	1.32
0.7	0.49	2.3	1.17	3.9	1.32
0.9	0.60	2.5	1.22	4.1	1.32
1.1	0.71	2.7	1.25	4.3	1.32
1.3	0.81	2.9	1.28	4.5	1.32

- Find a formula for the gorilla population in terms of t .
 - How long will it take for the gorilla population to reach the carrying capacity of the preserve?
- 7. Pacific halibut fishery** The Pacific halibut fishery has been modeled by the logistic equation

$$\frac{dy}{dt} = r(M - y)y$$

where $y(t)$ is the total weight of the halibut population in kilograms at time t (measured in years), the carrying capacity is estimated to be $M = 8 \times 10^7$ kg, and $r = 0.08875 \times 10^{-7}$ per year.

- If $y(0) = 1.6 \times 10^7$ kg, what is the total weight of the halibut population after 1 year?
 - When will the total weight in the halibut fishery reach 4×10^7 kg?
- 8. Modified logistic model** Suppose that the logistic differential equation in Example 2 is modified to

$$\frac{dP}{dt} = 0.001(100 - P)P - c$$

for some constant c .

- Explain the meaning of the constant c . What values for c might be realistic for the grizzly bear population?
- T** **b.** Draw a direction field for the differential equation when $c = 1$. What are the equilibrium solutions (Section 9.4)?
- Sketch several solution curves in your direction field from part (a). Describe what happens to the grizzly bear population for various initial populations.
- 9. Exact solutions** Find the exact solutions to the following initial value problems.
- $y' = 1 + y$, $y(0) = 1$
 - $y' = 0.5(400 - y)y$, $y(0) = 2$
- 10. Logistic differential equation** Show that the solution of the differential equation

$$\frac{dP}{dt} = r(M - P)P$$

is

$$P = \frac{M}{1 + Ae^{-rMt}},$$

where A is an arbitrary constant.

- 11. Catastrophic solution** Let k and P_0 be positive constants.

- Solve the initial value problem?

$$\frac{dP}{dt} = kP^2, \quad P(0) = P_0$$

- T** **b.** Show that the graph of the solution in part (a) has a vertical asymptote at a positive value of t . What is that value of t ?

- 12. Extinct populations** Consider the population model

$$\frac{dP}{dt} = r(M - P)(P - m),$$

where $r > 0$, M is the maximum sustainable population, and m is the minimum population below which the species becomes extinct.

- Let $m = 100$, and $M = 1200$, and assume that $m < P < M$. Show that the differential equation can be rewritten in the form

$$\left[\frac{1}{1200 - P} + \frac{1}{P - 100} \right] \frac{dP}{dt} = 1100r$$

and solve this separable equation.

- Find the solution to part (a) that satisfies $P(0) = 300$.
- Solve the differential equation with the restriction $m < P < M$.

Orthogonal Trajectories

In Exercises 13–18, find the orthogonal trajectories of the family of curves. Sketch several members of each family.

- $y = mx$
- $ky^2 + y^2 = 1$
- $y = ce^{-x}$
- $y = cx^2$
- $2x^2 + y^2 = c^2$
- $y = e^{kx}$
- Show that the curves $2x^2 + 3y^2 = 5$ and $y^2 = x^3$ are orthogonal.
- Find the family of solutions of the given differential equation and the family of orthogonal trajectories. Sketch both families.
 - $x dx + y dy = 0$
 - $x dy - 2y dx = 0$
- Suppose a and b are positive numbers. Sketch the parabolas

$$y^2 = 4a^2 - 4ax \quad \text{and} \quad y^2 = 4b^2 + 4bx$$

in the same diagram. Show that they intersect at $(a - b, \pm 2\sqrt{ab})$, and that each “ a -parabola” is orthogonal to every “ b -parabola.”

Chapter 9

Additional and Advanced Exercises

Theory and Applications

- 1. Transport through a cell membrane** Under some conditions, the result of the movement of a dissolved substance across a cell's membrane is described by the equation

$$\frac{dy}{dt} = k \frac{A}{V} (c - y).$$

In this equation, y is the concentration of the substance inside the cell and dy/dt is the rate at which y changes over time. The letters

k , A , V , and c stand for constants, k being the *permeability coefficient* (a property of the membrane), A the surface area of the membrane, V the cell's volume, and c the concentration of the substance outside the cell. The equation says that the rate at which the concentration changes within the cell is proportional to the difference between it and the outside concentration.

- a. Solve the equation for $y(t)$, using y_0 to denote $y(0)$.
 - b. Find the steady-state concentration, $\lim_{t \rightarrow \infty} y(t)$. (Based on *Some Mathematical Models in Biology*, edited by R. M. Thrall, J. A. Mortimer, K. R. Rebman, and R. F. Baum, rev. ed., Dec. 1967, PB-202 364, pp. 101–103; distributed by N.T.I.S., U.S. Department of Commerce.)
2. **Oxygen flow mixture** Oxygen flows through one tube into a liter flask filled with air, and the mixture of oxygen and air (considered well stirred) escapes through another tube. Assuming that air contains 21% oxygen, what percentage of oxygen will the flask contain after 5 L have passed through the intake tube?
3. **Carbon dioxide in a classroom** If the average person breathes 20 times per minute, exhaling each time 100 in^3 of air containing 4% carbon dioxide, find the percentage of carbon dioxide in the air of a $10,000 \text{ ft}^3$ closed room 1 hour after a class of 30 students enters. Assume that the air is fresh at the start, that the ventilators admit 1000 ft^3 of fresh air per minute, and that the fresh air contains 0.04% carbon dioxide.
4. **Height of a rocket** If an external force F acts upon a system whose mass varies with time, Newton's law of motion is

$$\frac{d(mv)}{dt} = F + (v + u) \frac{dm}{dt}.$$

In this equation, m is the mass of the system at time t , v its velocity, and $v + u$ is the velocity of the mass that is entering (or leaving) the system at the rate dm/dt . Suppose that a rocket of initial mass m_0 starts from rest, but is driven upward by firing some of its mass directly backward at the constant rate of $dm/dt = -b$ units per second and at constant speed relative to the rocket

$u = -c$. The only external force acting on the rocket is $F = -mg$ due to gravity. Under these assumptions, show that the height of the rocket above the ground at the end of t seconds (t small compared to m_0/b) is

$$y = c \left[t + \frac{m_0 - bt}{b} \ln \frac{m_0 - bt}{m_0} \right] - \frac{1}{2} g t^2.$$

5. a. Assume that $P(x)$ and $Q(x)$ are continuous over the interval $[a, b]$. Use the Fundamental Theorem of Calculus, Part 1 to show that any function y satisfying the equation

$$v(x)y = \int v(x)Q(x) dx + C$$

for $v(x) = e^{\int P(x) dx}$ is a solution to the first-order linear equation

$$\frac{dy}{dx} + P(x)y = Q(x).$$

- b. If $C = y_0 v(x_0) - \int_{x_0}^x v(t)Q(t) dt$, then show that any solution y in part (a) satisfies the initial condition $y(x_0) = y_0$.
6. (*Continuation of Exercise 5.*) Assume the hypotheses of Exercise 5, and assume that $y_1(x)$ and $y_2(x)$ are both solutions to the first-order linear equation satisfying the initial condition $y(x_0) = y_0$.
- a. Verify that $y(x) = y_1(x) - y_2(x)$ satisfies the initial value problem

$$y' + P(x)y = 0, \quad y(x_0) = 0.$$

- b. For the integrating factor $v(x) = e^{\int P(x) dx}$, show that

$$\frac{d}{dx} (v(x)[y_1(x) - y_2(x)]) = 0.$$

Conclude that $v(x)[y_1(x) - y_2(x)] = \text{constant}$.

- c. From part (a), we have $y_1(x_0) - y_2(x_0) = 0$. Since $v(x) > 0$ for $a < x < b$, use part (b) to establish that $y_1(x) - y_2(x) = 0$ on the interval (a, b) . Thus $y_1(x) = y_2(x)$ for all $a < x < b$.

Chapter 9

Practice Exercises

In Exercises 1–20 solve the differential equation.

1. $\frac{dy}{dx} = \sqrt{y} \cos^2 \sqrt{y}$
2. $y' = \frac{3y(x+1)^2}{y-1}$
3. $yy' = \sec y^2 \sec^2 x$
4. $y \cos^2 x \, dy + \sin x \, dx = 0$
5. $y' = xe^y \sqrt{x-2}$
6. $y' = xy e^{x^2}$
7. $\sec x \, dy + x \cos^2 y \, dx = 0$
8. $2x^2 \, dx - 3\sqrt{y} \csc x \, dy = 0$
9. $y' = \frac{e^y}{xy}$
10. $y' = xe^{x-y} \csc y$
11. $x(x-1) \, dy - y \, dx = 0$
12. $y' = (y^2 - 1)x^{-1}$
13. $2y' - y = xe^{x/2}$
14. $\frac{y'}{2} + y = e^{-x} \sin x$
15. $xy' + 2y = 1 - x^{-1}$
16. $xy' - y = 2x \ln x$
17. $(1 + e^x) \, dy + (ye^x + e^{-x}) \, dx = 0$
18. $e^{-x} \, dy + (e^{-x}y - 4x) \, dx = 0$
19. $(x + 3y^2) \, dy + y \, dx = 0$ (Hint: $d(xy) = y \, dx + x \, dy$)
20. $x \, dy + (3y - x^{-2} \cos x) \, dx = 0, \quad x > 0$

Initial Value Problems

In Exercises 21–30 solve the initial value problem.

21. $\frac{dy}{dx} = e^{-x-y-2}, \quad y(0) = -2$
22. $\frac{dy}{dx} = \frac{y \ln y}{1 + x^2}, \quad y(0) = e^2$
23. $(x+1) \frac{dy}{dx} + 2y = x, \quad x > -1, \quad y(0) = 1$

24. $x \frac{dy}{dx} + 2y = x^2 + 1, \quad x > 0, \quad y(1) = 1$
25. $\frac{dy}{dx} + 3x^2y = x^2, \quad y(0) = -1$
26. $x \, dy + (y - \cos x) \, dx = 0, \quad y\left(\frac{\pi}{2}\right) = 0$
27. $x \, dy - (y + \sqrt{y}) \, dx = 0, \quad y(1) = 1$
28. $y^{-2} \frac{dx}{dy} = \frac{e^x}{e^{2x} + 1}, \quad y(0) = 1$
29. $xy' + (x-2)y = 3x^3e^{-x}, \quad y(1) = 0$
30. $y \, dx + (3x - xy + 2) \, dy = 0, \quad y(2) = -1, \quad y < 0$

Euler's Method

In Exercises 31 and 32, use the stated method to solve the initial value problem on the given interval starting at x_0 with $dx = 0.1$.

- T** 31. **Euler:** $y' = y + \cos x, \quad y(0) = 0; \quad 0 \leq x \leq 2; \quad x_0 = 0$
- T** 32. **Improved Euler:** $y' = (2-y)(2x+3), \quad y(-3) = 1; \quad -3 \leq x \leq -1; \quad x_0 = -3$

In Exercises 33 and 34, use the stated method with $dx = 0.05$ to estimate $y(c)$ where y is the solution to the given initial value problem.

- T** 33. **Improved Euler:**

$$c = 3; \quad \frac{dy}{dx} = \frac{x-2y}{x+1}, \quad y(0) = 1$$

T 34. Euler:

$$c = 4; \quad \frac{dy}{dx} = \frac{x^2 - 2y + 1}{x}, \quad y(1) = 1$$

In Exercises 35 and 36, use the stated method to solve the initial value problem graphically, starting at $x_0 = 0$ with

- a. $dx = 0.1$. b. $dx = -0.1$.

T 35. Euler:

$$\frac{dy}{dx} = \frac{1}{e^{x+y+2}}, \quad y(0) = -2$$

T 36. Improved Euler:

$$\frac{dy}{dx} = -\frac{x^2 + y}{e^y + x}, \quad y(0) = 0$$

Slope Fields

In Exercises 37–40, sketch part of the equation's slope field. Then add to your sketch the solution curve that passes through the point $P(1, -1)$. Use Euler's method with $x_0 = 1$ and $dx = 0.2$ to estimate $y(2)$. Round your answers to four decimal places. Find the exact value of $y(2)$ for comparison.

37. $y' = x$ 38. $y' = 1/x$

39. $y' = xy$ 40. $y' = 1/y$

Autonomous Differential Equations and Phase Lines

In Exercises 41 and 42.

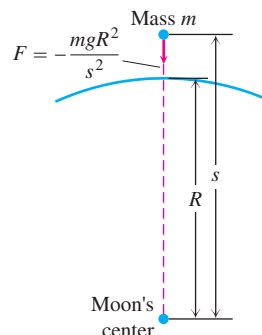
- Identify the equilibrium values. Which are stable and which are unstable?
- Construct a phase line. Identify the signs of y' and y'' .
- Sketch a representative selection of solution curves.

41. $\frac{dy}{dx} = y^2 - 1$

42. $\frac{dy}{dx} = y - y^2$

Applications

- 43. Escape velocity** The gravitational attraction F exerted by an airless moon on a body of mass m at a distance s from the moon's center is given by the equation $F = -mgR^2s^{-2}$, where g is the acceleration of gravity at the moon's surface and R is the moon's radius (see accompanying figure). The force F is negative because it acts in the direction of decreasing s .



- a. If the body is projected vertically upward from the moon's surface with an initial velocity v_0 at time $t = 0$, use Newton's second law, $F = ma$, to show that the body's velocity at position s is given by the equation

$$v^2 = \frac{2gR^2}{s} + v_0^2 - 2gR.$$

Thus, the velocity remains positive as long as $v_0 \geq \sqrt{2gR}$.

The velocity $v_0 = \sqrt{2gR}$ is the moon's **escape velocity**. A body projected upward with this velocity or a greater one will escape from the moon's gravitational pull.

- b. Show that if $v_0 = \sqrt{2gR}$, then

$$s = R \left(1 + \frac{3v_0}{2R} t \right)^{2/3}.$$

- 44. Coasting to a stop** Table 9.9 shows the distance s (meters) coasted on in-line skates in t sec by Johnathon Krueger. Find a model for his position in the form of Equation (2) of Section 9.5. His initial velocity was $v_0 = 0.86$ m/sec, his mass $m = 30.84$ kg (he weighed 68 lb), and his total coasting distance 0.97 m.

TABLE 9.9 Johnathon Krueger skating data

t (sec)	s (m)	t (sec)	s (m)	t (sec)	s (m)
0	0	0.93	0.61	1.86	0.93
0.13	0.08	1.06	0.68	2.00	0.94
0.27	0.19	1.20	0.74	2.13	0.95
0.40	0.28	1.33	0.79	2.26	0.96
0.53	0.36	1.46	0.83	2.39	0.96
0.67	0.45	1.60	0.87	2.53	0.97
0.80	0.53	1.73	0.90	2.66	0.97

Chapter 9

Questions to Guide Your Review

1. What is a first-order differential equation? When is a function a solution of such an equation?
2. How do you solve separable first-order differential equations?
3. What is the law of exponential change? How can it be derived from an initial value problem? What are some of the applications of the law?
4. What is the slope field of a differential equation $y' = f(x, y)$? What can we learn from such fields?
5. How do you solve linear first-order differential equations?
6. Describe Euler's method for solving the initial value problem $y' = f(x, y), y(x_0) = y_0$ numerically. Give an example. Comment on the method's accuracy. Why might you want to solve an initial value problem numerically?
7. Describe the improved Euler's method for solving the initial value problem $y' = f(x, y), y(x_0) = y_0$ numerically. How does it compare with Euler's method?
8. What is an autonomous differential equation? What are its equilibrium values? How do they differ from critical points? What is a stable equilibrium value? Unstable?
9. How do you construct the phase line for an autonomous differential equation? How does the phase line help you produce a graph which qualitatively depicts a solution to the differential equation?
10. Why is the exponential model unrealistic for predicting long-term population growth? How does the logistic model correct for the deficiency in the exponential model for population growth? What is the logistic differential equation? What is the form of its solution? Describe the graph of the logistic solution.

Chapter 9 Technology Application Projects

Mathematica/Maple Module

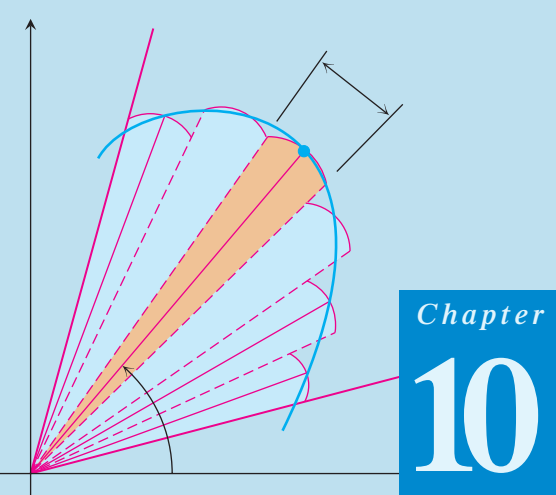
Drug Dosages: Are They Effective? Are They Safe?

Formulate and solve an initial value model for the absorption of blood in the bloodstream.

Mathematica/Maple Module

First-Order Differential Equations and Slope Fields

Plot slope fields and solution curves for various initial conditions to selected first-order differential equations.



Chapter

10

CONIC SECTIONS AND POLAR COORDINATES

OVERVIEW In this chapter we give geometric definitions of parabolas, ellipses, and hyperbolas and derive their standard equations. These curves are called *conic sections*, or *conics*, and model the paths traveled by planets, satellites, and other bodies whose motions are driven by inverse square forces. In Chapter 13 we will see that once the path of a moving body is known to be a conic, we immediately have information about the body's velocity and the force that drives it. Planetary motion is best described with the help of polar coordinates, so we also investigate curves, derivatives, and integrals in this new coordinate system.

10.1

Conic Sections and Quadratic Equations

In Chapter 1 we defined a **circle** as the set of points in a plane whose distance from some fixed center point is a constant radius value. If the center is (h, k) and the radius is a , the standard equation for the circle is $(x - h)^2 + (y - k)^2 = a^2$. It is an example of a conic section, which are the curves formed by cutting a double cone with a plane (Figure 10.1); hence the name *conic section*.

We now describe parabolas, ellipses, and hyperbolas as the graphs of quadratic equations in the coordinate plane.

Parabolas

DEFINITIONS Parabola, Focus, Directrix

A set that consists of all the points in a plane equidistant from a given fixed point and a given fixed line in the plane is a **parabola**. The fixed point is the **focus** of the parabola. The fixed line is the **directrix**.

If the focus F lies on the directrix L , the parabola is the line through F perpendicular to L . We consider this to be a degenerate case and assume henceforth that F does not lie on L .

A parabola has its simplest equation when its focus and directrix straddle one of the coordinate axes. For example, suppose that the focus lies at the point $F(0, p)$ on the positive y -axis and that the directrix is the line $y = -p$ (Figure 10.2). In the notation of the figure,

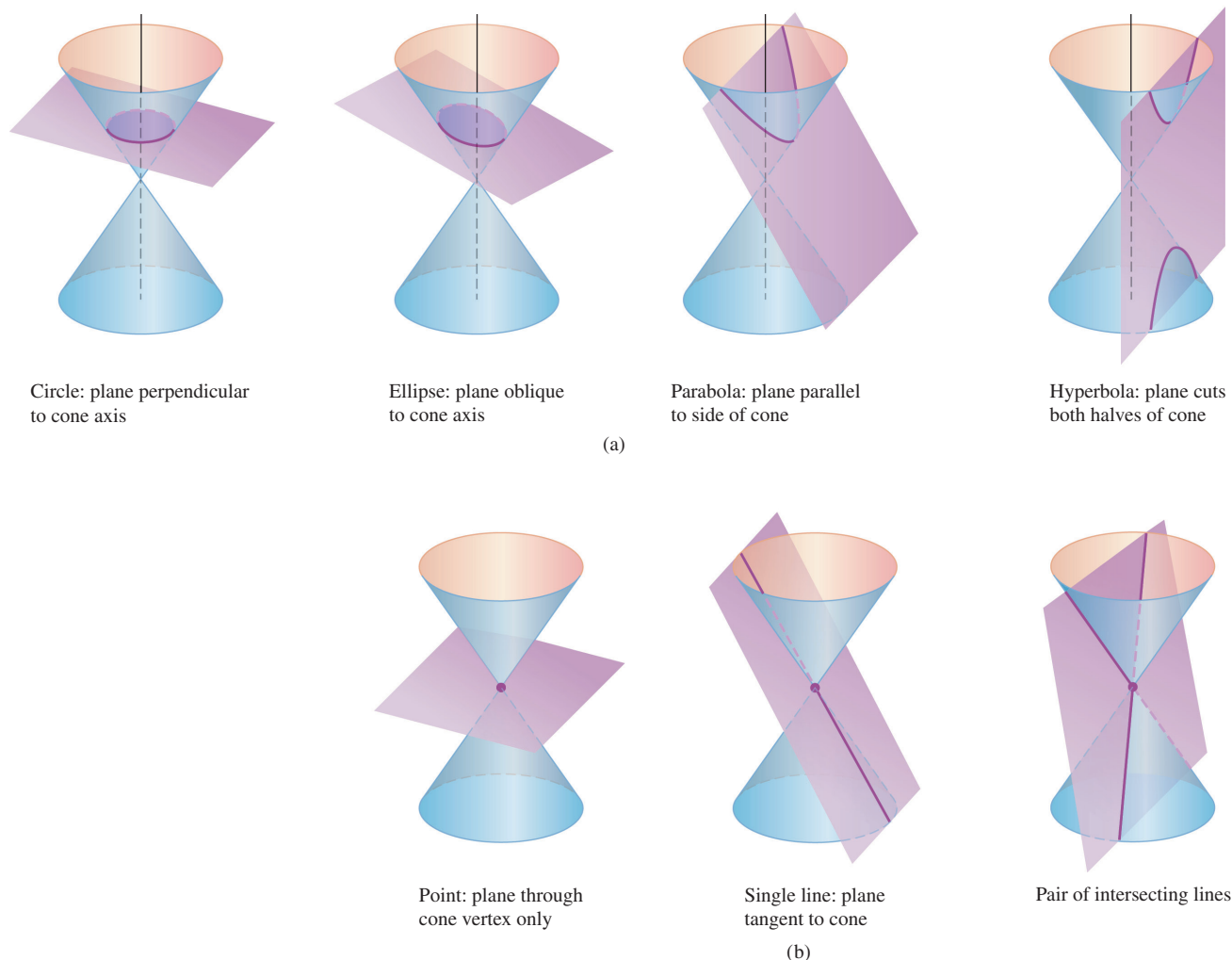


FIGURE 10.1 The standard conic sections (a) are the curves in which a plane cuts a double cone. Hyperbolas come in two parts, called *branches*. The point and lines obtained by passing the plane through the cone's vertex (b) are *degenerate* conic sections.

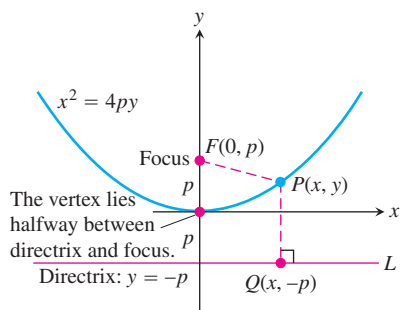


FIGURE 10.2 The standard form of the parabola $x^2 = 4py$, $p > 0$.

a point $P(x, y)$ lies on the parabola if and only if $PF = PQ$. From the distance formula,

$$PF = \sqrt{(x - 0)^2 + (y - p)^2} = \sqrt{x^2 + (y - p)^2}$$

$$PQ = \sqrt{(x - x)^2 + (y - (-p))^2} = \sqrt{(y + p)^2}.$$

When we equate these expressions, square, and simplify, we get

$$y = \frac{x^2}{4p} \quad \text{or} \quad x^2 = 4py. \quad \text{Standard form} \quad (1)$$

These equations reveal the parabola's symmetry about the y -axis. We call the y -axis the **axis** of the parabola (short for "axis of symmetry").

The point where a parabola crosses its axis is the **vertex**. The vertex of the parabola $x^2 = 4py$ lies at the origin (Figure 10.2). The positive number p is the parabola's **focal length**.

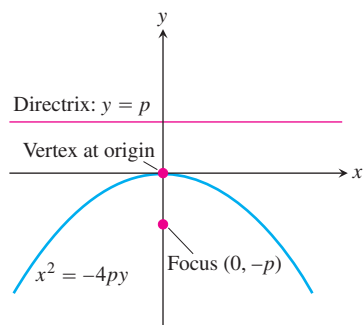


FIGURE 10.3 The parabola $x^2 = -4py$, $p > 0$.

If the parabola opens downward, with its focus at $(0, -p)$ and its directrix the line $y = p$, then Equations (1) become

$$y = -\frac{x^2}{4p} \quad \text{and} \quad x^2 = -4py$$

(Figure 10.3). We obtain similar equations for parabolas opening to the right or to the left (Figure 10.4 and Table 10.1).

TABLE 10.1 Standard-form equations for parabolas with vertices at the origin ($p > 0$)

Equation	Focus	Directrix	Axis	Opens
$x^2 = 4py$	$(0, p)$	$y = -p$	y -axis	Up
$x^2 = -4py$	$(0, -p)$	$y = p$	y -axis	Down
$y^2 = 4px$	$(p, 0)$	$x = -p$	x -axis	To the right
$y^2 = -4px$	$(-p, 0)$	$x = p$	x -axis	To the left

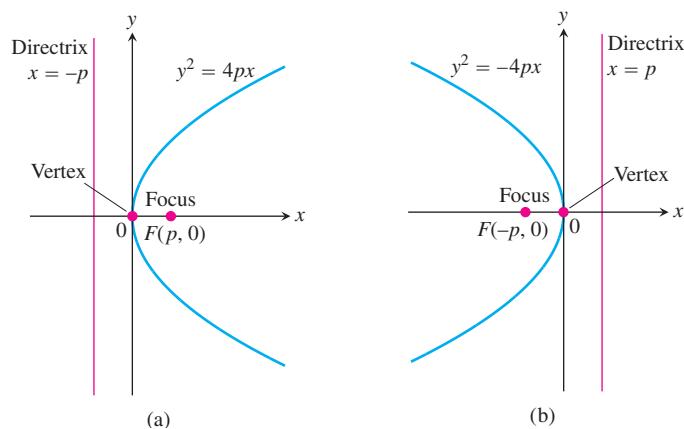


FIGURE 10.4 (a) The parabola $y^2 = 4px$. (b) The parabola $y^2 = -4px$.

EXAMPLE 1 Find the focus and directrix of the parabola $y^2 = 10x$.

Solution We find the value of p in the standard equation $y^2 = 4px$:

$$4p = 10, \quad \text{so} \quad p = \frac{10}{4} = \frac{5}{2}.$$

Then we find the focus and directrix for this value of p :

$$\text{Focus:} \quad (p, 0) = \left(\frac{5}{2}, 0\right)$$

$$\text{Directrix:} \quad x = -p \quad \text{or} \quad x = -\frac{5}{2}.$$



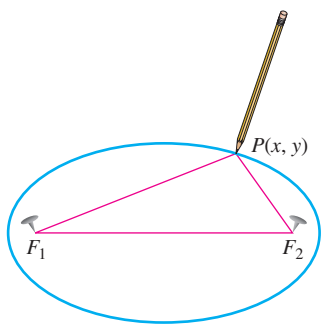


FIGURE 10.5 One way to draw an ellipse uses two tacks and a loop of string to guide the pencil.

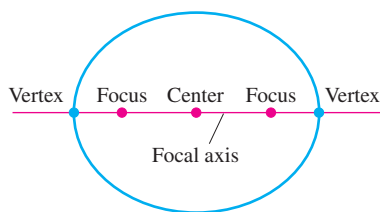


FIGURE 10.6 Points on the focal axis of an ellipse.

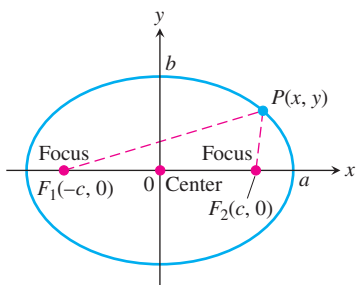


FIGURE 10.7 The ellipse defined by the equation $PF_1 + PF_2 = 2a$ is the graph of the equation $(x^2/a^2) + (y^2/b^2) = 1$, where $b^2 = a^2 - c^2$.

The horizontal and vertical shift formulas in Section 1.5, can be applied to the equations in Table 10.1 to give equations for a variety of parabolas in other locations (see Exercises 39, 40, and 45–48).

Ellipses

DEFINITIONS Ellipse, Foci

An **ellipse** is the set of points in a plane whose distances from two fixed points in the plane have a constant sum. The two fixed points are the **foci** of the ellipse.

The quickest way to construct an ellipse uses the definition. Put a loop of string around two tacks F_1 and F_2 , pull the string taut with a pencil point P , and move the pencil around to trace a closed curve (Figure 10.5). The curve is an ellipse because the sum $PF_1 + PF_2$, being the length of the loop minus the distance between the tacks, remains constant. The ellipse's foci lie at F_1 and F_2 .

DEFINITIONS Focal Axis, Center, Vertices

The line through the foci of an ellipse is the ellipse's **focal axis**. The point on the axis halfway between the foci is the **center**. The points where the focal axis and ellipse cross are the ellipse's **vertices** (Figure 10.6).

If the foci are $F_1(-c, 0)$ and $F_2(c, 0)$ (Figure 10.7), and $PF_1 + PF_2$ is denoted by $2a$, then the coordinates of a point P on the ellipse satisfy the equation

$$\sqrt{(x+c)^2 + y^2} + \sqrt{(x-c)^2 + y^2} = 2a.$$

To simplify this equation, we move the second radical to the right-hand side, square, isolate the remaining radical, and square again, obtaining

$$\frac{x^2}{a^2} + \frac{y^2}{a^2 - c^2} = 1. \quad (2)$$

Since $PF_1 + PF_2$ is greater than the length F_1F_2 (triangle inequality for triangle PF_1F_2), the number $2a$ is greater than $2c$. Accordingly, $a > c$ and the number $a^2 - c^2$ in Equation (2) is positive.

The algebraic steps leading to Equation (2) can be reversed to show that every point P whose coordinates satisfy an equation of this form with $0 < c < a$ also satisfies the equation $PF_1 + PF_2 = 2a$. A point therefore lies on the ellipse if and only if its coordinates satisfy Equation (2).

If

$$b = \sqrt{a^2 - c^2}, \quad (3)$$

then $a^2 - c^2 = b^2$ and Equation (2) takes the form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1. \quad (4)$$

Equation (4) reveals that this ellipse is symmetric with respect to the origin and both coordinate axes. It lies inside the rectangle bounded by the lines $x = \pm a$ and $y = \pm b$. It crosses the axes at the points $(\pm a, 0)$ and $(0, \pm b)$. The tangents at these points are perpendicular to the axes because

$$\frac{dy}{dx} = -\frac{b^2x}{a^2y} \quad \begin{array}{l} \text{Obtained from Equation (4)} \\ \text{by implicit differentiation} \end{array}$$

is zero if $x = 0$ and infinite if $y = 0$.

The **major axis** of the ellipse in Equation (4) is the line segment of length $2a$ joining the points $(\pm a, 0)$. The **minor axis** is the line segment of length $2b$ joining the points $(0, \pm b)$. The number a itself is the **semimajor axis**, the number b the **semiminor axis**. The number c , found from Equation (3) as

$$c = \sqrt{a^2 - b^2},$$

is the **center-to-focus distance** of the ellipse.

EXAMPLE 2 Major Axis Horizontal

The ellipse

$$\frac{x^2}{16} + \frac{y^2}{9} = 1 \quad (5)$$

(Figure 10.8) has

$$\text{Semimajor axis: } a = \sqrt{16} = 4, \quad \text{Semiminor axis: } b = \sqrt{9} = 3$$

$$\text{Center-to-focus distance: } c = \sqrt{16 - 9} = \sqrt{7}$$

$$\text{Foci: } (\pm c, 0) = (\pm\sqrt{7}, 0)$$

$$\text{Vertices: } (\pm a, 0) = (\pm 4, 0)$$

$$\text{Center: } (0, 0).$$

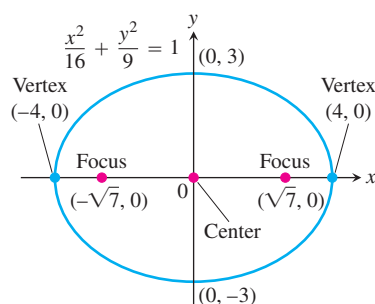


FIGURE 10.8 An ellipse with its major axis horizontal (Example 2).

EXAMPLE 3 Major Axis Vertical

The ellipse

$$\frac{x^2}{9} + \frac{y^2}{16} = 1, \quad (6)$$

obtained by interchanging x and y in Equation (5), has its major axis vertical instead of horizontal (Figure 10.9). With a^2 still equal to 16 and b^2 equal to 9, we have

$$\text{Semimajor axis: } a = \sqrt{16} = 4, \quad \text{Semiminor axis: } b = \sqrt{9} = 3$$

$$\text{Center-to-focus distance: } c = \sqrt{16 - 9} = \sqrt{7}$$

$$\text{Foci: } (0, \pm c) = (0, \pm\sqrt{7})$$

$$\text{Vertices: } (0, \pm a) = (0, \pm 4)$$

$$\text{Center: } (0, 0).$$

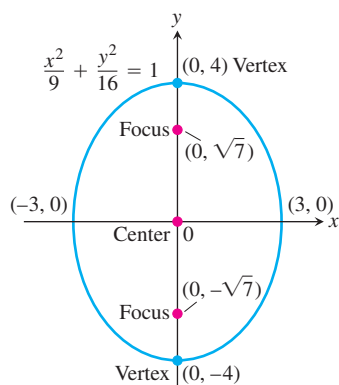


FIGURE 10.9 An ellipse with its major axis vertical (Example 3).

There is never any cause for confusion in analyzing Equations (5) and (6). We simply find the intercepts on the coordinate axes; then we know which way the major axis runs because it is the longer of the two axes. The center always lies at the origin and the foci and vertices lie on the major axis.

Standard-Form Equations for Ellipses Centered at the Origin

Foci on the x-axis: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (a > b)$

Center-to-focus distance: $c = \sqrt{a^2 - b^2}$

Foci: $(\pm c, 0)$

Vertices: $(\pm a, 0)$

Foci on the y-axis: $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1 \quad (a > b)$

Center-to-focus distance: $c = \sqrt{a^2 - b^2}$

Foci: $(0, \pm c)$

Vertices: $(0, \pm a)$

In each case, a is the semimajor axis and b is the semiminor axis.

Hyperbolas

DEFINITIONS Hyperbola, Foci

A **hyperbola** is the set of points in a plane whose distances from two fixed points in the plane have a constant difference. The two fixed points are the **foci** of the hyperbola.

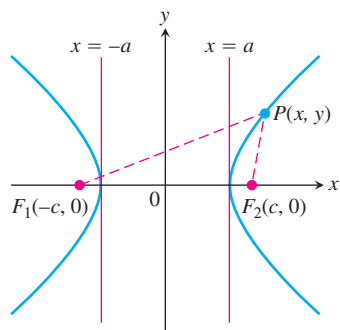


FIGURE 10.10 Hyperbolas have two branches. For points on the right-hand branch of the hyperbola shown here, $PF_1 - PF_2 = 2a$. For points on the left-hand branch, $PF_2 - PF_1 = 2a$. We then let $b = \sqrt{c^2 - a^2}$.

If the foci are $F_1(-c, 0)$ and $F_2(c, 0)$ (Figure 10.10) and the constant difference is $2a$, then a point (x, y) lies on the hyperbola if and only if

$$\sqrt{(x + c)^2 + y^2} - \sqrt{(x - c)^2 + y^2} = \pm 2a. \quad (7)$$

To simplify this equation, we move the second radical to the right-hand side, square, isolate the remaining radical, and square again, obtaining

$$\frac{x^2}{a^2} + \frac{y^2}{a^2 - c^2} = 1. \quad (8)$$

So far, this looks just like the equation for an ellipse. But now $a^2 - c^2$ is negative because $2a$, being the difference of two sides of triangle PF_1F_2 , is less than $2c$, the third side.

The algebraic steps leading to Equation (8) can be reversed to show that every point P whose coordinates satisfy an equation of this form with $0 < a < c$ also satisfies Equation (7). A point therefore lies on the hyperbola if and only if its coordinates satisfy Equation (8).

If we let b denote the positive square root of $c^2 - a^2$,

$$b = \sqrt{c^2 - a^2}, \quad (9)$$

then $a^2 - c^2 = -b^2$ and Equation (8) takes the more compact form

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1. \quad (10)$$

The differences between Equation (10) and the equation for an ellipse (Equation 4) are the minus sign and the new relation

$$c^2 = a^2 + b^2. \quad \text{From Equation (9)}$$

Like the ellipse, the hyperbola is symmetric with respect to the origin and coordinate axes. It crosses the x -axis at the points $(\pm a, 0)$. The tangents at these points are vertical because

$$\frac{dy}{dx} = \frac{b^2 x}{a^2 y} \quad \begin{array}{l} \text{Obtained from Equation (10)} \\ \text{by implicit differentiation} \end{array}$$

is infinite when $y = 0$. The hyperbola has no y -intercepts; in fact, no part of the curve lies between the lines $x = -a$ and $x = a$.

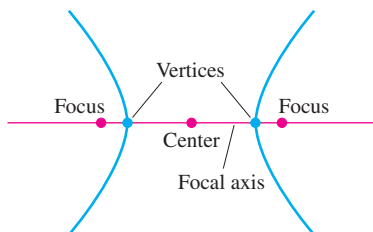


FIGURE 10.11 Points on the focal axis of a hyperbola.

DEFINITIONS Focal Axis, Center, Vertices

The line through the foci of a hyperbola is the **focal axis**. The point on the axis halfway between the foci is the hyperbola's **center**. The points where the focal axis and hyperbola cross are the **vertices** (Figure 10.11).

Asymptotes of Hyperbolas and Graphing

If we solve Equation (10) for y we obtain

$$\begin{aligned} y^2 &= b^2 \left(\frac{x^2}{a^2} - 1 \right) \\ &= \frac{b^2}{a^2} x^2 \left(1 - \frac{a^2}{x^2} \right) \end{aligned}$$

or, taking square roots,

$$y = \pm \frac{b}{a} x \sqrt{1 - \frac{a^2}{x^2}}.$$

As $x \rightarrow \pm\infty$, the factor $\sqrt{1 - a^2/x^2}$ approaches 1, and the factor $\pm(b/a)x$ is dominant. Thus the lines

$$y = \pm \frac{b}{a} x$$

are the two **asymptotes** of the hyperbola defined by Equation (10). The asymptotes give the guidance we need to graph hyperbolas quickly. The fastest way to find the equations of the asymptotes is to replace the 1 in Equation (10) by 0 and solve the new equation for y :

$$\underbrace{\frac{x^2}{a^2} - \frac{y^2}{b^2}}_{\text{hyperbola}} = 1 \rightarrow \underbrace{\frac{x^2}{a^2} - \frac{y^2}{b^2}}_{0 \text{ for } 1} = 0 \rightarrow \underbrace{y = \pm \frac{b}{a} x}_{\text{asymptotes}}.$$

Standard-Form Equations for Hyperbolas Centered at the Origin

Foci on the x -axis: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Center-to-focus distance: $c = \sqrt{a^2 + b^2}$

Foci: $(\pm c, 0)$

Vertices: $(\pm a, 0)$

Asymptotes: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$ or $y = \pm \frac{b}{a}x$

Notice the difference in the asymptote equations (b/a in the first, a/b in the second).

Foci on the y -axis: $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$

Center-to-focus distance: $c = \sqrt{a^2 + b^2}$

Foci: $(0, \pm c)$

Vertices: $(0, \pm a)$

Asymptotes: $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 0$ or $y = \pm \frac{a}{b}x$

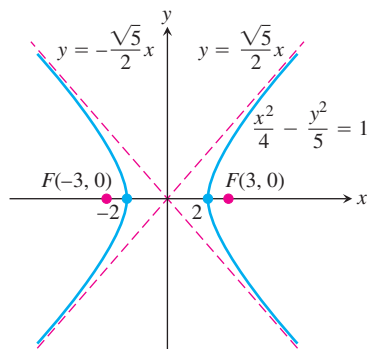


FIGURE 10.12 The hyperbola and its asymptotes in Example 4.

EXAMPLE 4 Foci on the x -axis

The equation

$$\frac{x^2}{4} - \frac{y^2}{5} = 1 \quad (11)$$

is Equation (10) with $a^2 = 4$ and $b^2 = 5$ (Figure 10.12). We have

Center-to-focus distance: $c = \sqrt{a^2 + b^2} = \sqrt{4 + 5} = 3$

Foci: $(\pm c, 0) = (\pm 3, 0)$, Vertices: $(\pm a, 0) = (\pm 2, 0)$

Center: $(0, 0)$

Asymptotes: $\frac{x^2}{4} - \frac{y^2}{5} = 0$ or $y = \pm \frac{\sqrt{5}}{2}x$. ■

EXAMPLE 5 Foci on the y -axis

The hyperbola

$$\frac{y^2}{4} - \frac{x^2}{5} = 1,$$

obtained by interchanging x and y in Equation (11), has its vertices on the y -axis instead of the x -axis (Figure 10.13). With a^2 still equal to 4 and b^2 equal to 5, we have

Center-to-focus distance: $c = \sqrt{a^2 + b^2} = \sqrt{4 + 5} = 3$

Foci: $(0, \pm c) = (0, \pm 3)$, Vertices: $(0, \pm a) = (0, \pm 2)$

Center: $(0, 0)$

Asymptotes: $\frac{y^2}{4} - \frac{x^2}{5} = 0$ or $y = \pm \frac{2}{\sqrt{5}}x$. ■

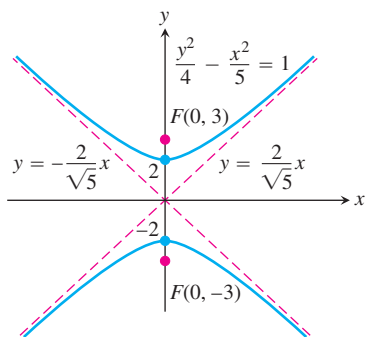


FIGURE 10.13 The hyperbola and its asymptotes in Example 5.

Reflective Properties

The chief applications of parabolas involve their use as reflectors of light and radio waves. Rays originating at a parabola's focus are reflected out of the parabola parallel to the parabola's axis (Figure 10.14 and Exercise 90). Moreover, the time any ray takes from the focus to a line parallel to the parabola's directrix (thus perpendicular to its axis) is the same for each of the rays. These properties are used by flashlight, headlight, and spotlight reflectors and by microwave broadcast antennas.

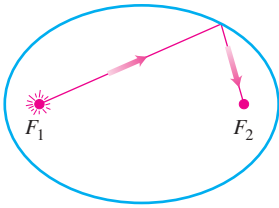


FIGURE 10.15 An elliptical mirror (shown here in profile) reflects light from one focus to the other.

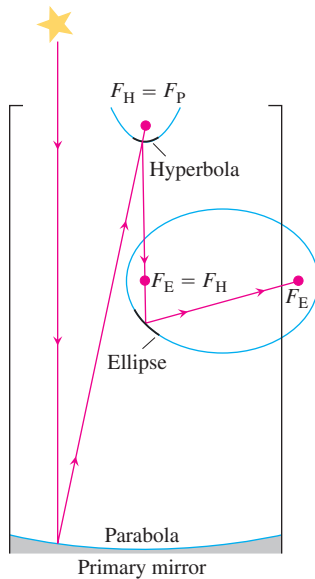


FIGURE 10.16 Schematic drawing of a reflecting telescope.

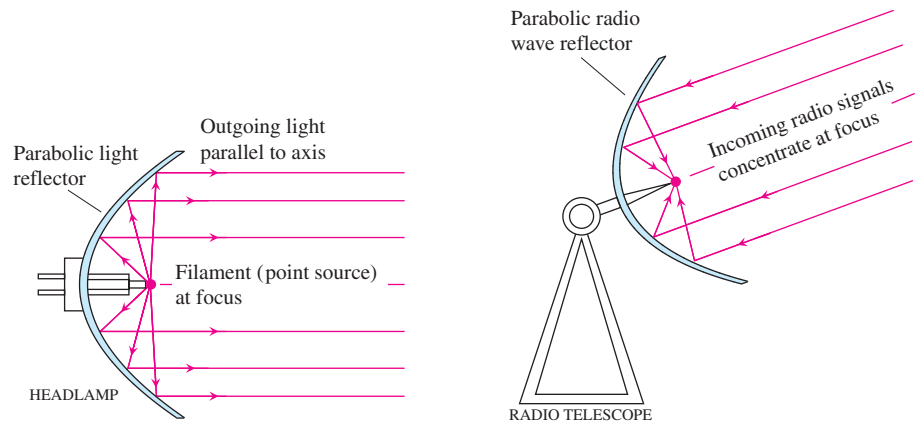


FIGURE 10.14 Parabolic reflectors can generate a beam of light parallel to the parabola's axis from a source at the focus; or they can receive rays parallel to the axis and concentrate them at the focus.

If an ellipse is revolved about its major axis to generate a surface (the surface is called an *ellipsoid*) and the interior is silvered to produce a mirror, light from one focus will be reflected to the other focus (Figure 10.15). Ellipsoids reflect sound the same way, and this property is used to construct *whispering galleries*, rooms in which a person standing at one focus can hear a whisper from the other focus. (Statuary Hall in the U.S. Capitol building is a whispering gallery.)

Light directed toward one focus of a hyperbolic mirror is reflected toward the other focus. This property of hyperbolas is combined with the reflective properties of parabolas and ellipses in designing some modern telescopes. In Figure 10.16 starlight reflects off a primary parabolic mirror toward the mirror's focus F_P . It is then reflected by a small hyperbolic mirror, whose focus is $F_H = F_P$, toward the second focus of the hyperbola, $F_E = F_H$. Since this focus is shared by an ellipse, the light is reflected by the elliptical mirror to the ellipse's second focus to be seen by an observer.

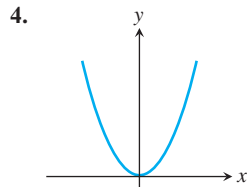
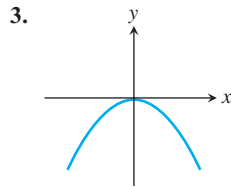
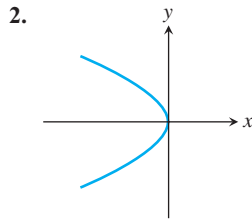
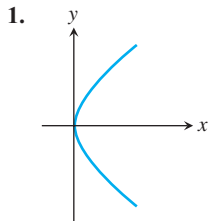
EXERCISES 10.1

Identifying Graphs

Match the parabolas in Exercises 1–4 with the following equations:

$$x^2 = 2y, \quad x^2 = -6y, \quad y^2 = 8x, \quad y^2 = -4x.$$

Then find the parabola's focus and directrix.

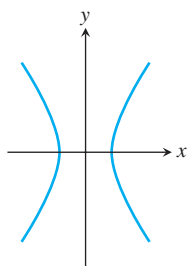


Match each conic section in Exercises 5–8 with one of these equations:

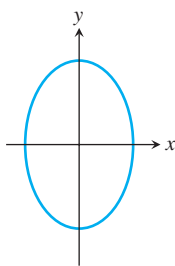
$$\begin{aligned} \frac{x^2}{4} + \frac{y^2}{9} &= 1, & \frac{x^2}{2} + y^2 &= 1, \\ \frac{y^2}{4} - x^2 &= 1, & \frac{x^2}{4} - \frac{y^2}{9} &= 1. \end{aligned}$$

Then find the conic section's foci and vertices. If the conic section is a hyperbola, find its asymptotes as well.

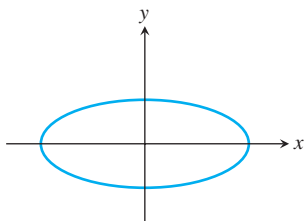
5.



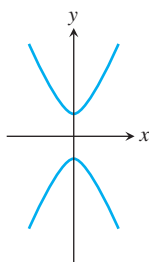
6.



7.



8.



Parabolas

Exercises 9–16 give equations of parabolas. Find each parabola's focus and directrix. Then sketch the parabola. Include the focus and directrix in your sketch.

- | | | |
|-----------------|----------------|-----------------|
| 9. $y^2 = 12x$ | 10. $x^2 = 6y$ | 11. $x^2 = -8y$ |
| 12. $y^2 = -2x$ | 13. $y = 4x^2$ | 14. $y = -8x^2$ |
| 15. $x = -3y^2$ | 16. $x = 2y^2$ | |

Ellipses

Exercises 17–24 give equations for ellipses. Put each equation in standard form. Then sketch the ellipse. Include the foci in your sketch.

- | | |
|---------------------------|-----------------------------|
| 17. $16x^2 + 25y^2 = 400$ | 18. $7x^2 + 16y^2 = 112$ |
| 19. $2x^2 + y^2 = 2$ | 20. $2x^2 + y^2 = 4$ |
| 21. $3x^2 + 2y^2 = 6$ | 22. $9x^2 + 10y^2 = 90$ |
| 23. $6x^2 + 9y^2 = 54$ | 24. $169x^2 + 25y^2 = 4225$ |

Exercises 25 and 26 give information about the foci and vertices of ellipses centered at the origin of the xy -plane. In each case, find the ellipse's standard-form equation from the given information.

- | | |
|------------------------------|------------------------|
| 25. Foci: $(\pm\sqrt{2}, 0)$ | 26. Foci: $(0, \pm 4)$ |
| Vertices: $(\pm 2, 0)$ | Vertices: $(0, \pm 5)$ |

Hyperbolas

Exercises 27–34 give equations for hyperbolas. Put each equation in standard form and find the hyperbola's asymptotes. Then sketch the hyperbola. Include the asymptotes and foci in your sketch.

- | | |
|---------------------|--------------------------|
| 27. $x^2 - y^2 = 1$ | 28. $9x^2 - 16y^2 = 144$ |
|---------------------|--------------------------|

- | | |
|------------------------|----------------------------|
| 29. $y^2 - x^2 = 8$ | 30. $y^2 - x^2 = 4$ |
| 31. $8x^2 - 2y^2 = 16$ | 32. $y^2 - 3x^2 = 3$ |
| 33. $8y^2 - 2x^2 = 16$ | 34. $64x^2 - 36y^2 = 2304$ |

Exercises 35–38 give information about the foci, vertices, and asymptotes of hyperbolas centered at the origin of the xy -plane. In each case, find the hyperbola's standard-form equation from the information given.

- | | |
|------------------------------------|---|
| 35. Foci: $(0, \pm\sqrt{2})$ | 36. Foci: $(\pm 2, 0)$ |
| Asymptotes: $y = \pm x$ | Asymptotes: $y = \pm \frac{1}{\sqrt{3}}x$ |
| 37. Vertices: $(\pm 3, 0)$ | 38. Vertices: $(0, \pm 2)$ |
| Asymptotes: $y = \pm \frac{4}{3}x$ | Asymptotes: $y = \pm \frac{1}{2}x$ |

Shifting Conic Sections

39. The parabola $y^2 = 8x$ is shifted down 2 units and right 1 unit to generate the parabola $(y + 2)^2 = 8(x - 1)$.
- Find the new parabola's vertex, focus, and directrix.
 - Plot the new vertex, focus, and directrix, and sketch in the parabola.
40. The parabola $x^2 = -4y$ is shifted left 1 unit and up 3 units to generate the parabola $(x + 1)^2 = -4(y - 3)$.
- Find the new parabola's vertex, focus, and directrix.
 - Plot the new vertex, focus, and directrix, and sketch in the parabola.
41. The ellipse $(x^2/16) + (y^2/9) = 1$ is shifted 4 units to the right and 3 units up to generate the ellipse

$$\frac{(x - 4)^2}{16} + \frac{(y - 3)^2}{9} = 1.$$

- Find the foci, vertices, and center of the new ellipse.
 - Plot the new foci, vertices, and center, and sketch in the new ellipse.
42. The ellipse $(x^2/9) + (y^2/25) = 1$ is shifted 3 units to the left and 2 units down to generate the ellipse

$$\frac{(x + 3)^2}{9} + \frac{(y + 2)^2}{25} = 1.$$

- Find the foci, vertices, and center of the new ellipse.
 - Plot the new foci, vertices, and center, and sketch in the new ellipse.
43. The hyperbola $(x^2/16) - (y^2/9) = 1$ is shifted 2 units to the right to generate the hyperbola

$$\frac{(x - 2)^2}{16} - \frac{y^2}{9} = 1.$$

- Find the center, foci, vertices, and asymptotes of the new hyperbola.

- b. Plot the new center, foci, vertices, and asymptotes, and sketch in the hyperbola.
44. The hyperbola $(y^2/4) - (x^2/5) = 1$ is shifted 2 units down to generate the hyperbola

$$\frac{(y + 2)^2}{4} - \frac{x^2}{5} = 1.$$

- a. Find the center, foci, vertices, and asymptotes of the new hyperbola.
- b. Plot the new center, foci, vertices, and asymptotes, and sketch in the hyperbola.

Exercises 45–48 give equations for parabolas and tell how many units up or down and to the right or left each parabola is to be shifted. Find an equation for the new parabola, and find the new vertex, focus, and directrix.

45. $y^2 = 4x$, left 2, down 3 46. $y^2 = -12x$, right 4, up 3
 47. $x^2 = 8y$, right 1, down 7 48. $x^2 = 6y$, left 3, down 2

Exercises 49–52 give equations for ellipses and tell how many units up or down and to the right or left each ellipse is to be shifted. Find an equation for the new ellipse, and find the new foci, vertices, and center.

49. $\frac{x^2}{6} + \frac{y^2}{9} = 1$, left 2, down 1
 50. $\frac{x^2}{2} + y^2 = 1$, right 3, up 4
 51. $\frac{x^2}{3} + \frac{y^2}{2} = 1$, right 2, up 3
 52. $\frac{x^2}{16} + \frac{y^2}{25} = 1$, left 4, down 5

Exercises 53–56 give equations for hyperbolas and tell how many units up or down and to the right or left each hyperbola is to be shifted. Find an equation for the new hyperbola, and find the new center, foci, vertices, and asymptotes.

53. $\frac{x^2}{4} - \frac{y^2}{5} = 1$, right 2, up 2
 54. $\frac{x^2}{16} - \frac{y^2}{9} = 1$, left 2, down 1
 55. $y^2 - x^2 = 1$, left 1, down 1
 56. $\frac{y^2}{3} - x^2 = 1$, right 1, up 3

Find the center, foci, vertices, asymptotes, and radius, as appropriate, of the conic sections in Exercises 57–68.

57. $x^2 + 4x + y^2 = 12$
 58. $2x^2 + 2y^2 - 28x + 12y + 114 = 0$
 59. $x^2 + 2x + 4y - 3 = 0$ 60. $y^2 - 4y - 8x - 12 = 0$
 61. $x^2 + 5y^2 + 4x = 1$ 62. $9x^2 + 6y^2 + 36y = 0$
 63. $x^2 + 2y^2 - 2x - 4y = -1$

64. $4x^2 + y^2 + 8x - 2y = -1$
 65. $x^2 - y^2 - 2x + 4y = 4$ 66. $x^2 - y^2 + 4x - 6y = 6$
 67. $2x^2 - y^2 + 6y = 3$ 68. $y^2 - 4x^2 + 16x = 24$

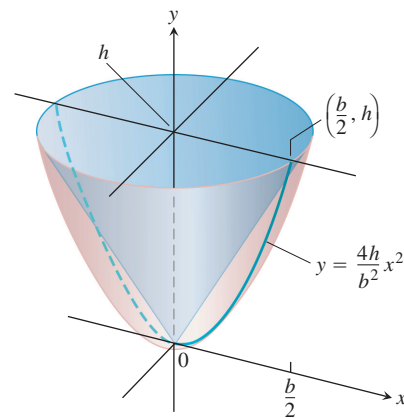
Inequalities

Sketch the regions in the xy -plane whose coordinates satisfy the inequalities or pairs of inequalities in Exercises 69–74.

69. $9x^2 + 16y^2 \leq 144$
 70. $x^2 + y^2 \geq 1$ and $4x^2 + y^2 \leq 4$
 71. $x^2 + 4y^2 \geq 4$ and $4x^2 + 9y^2 \leq 36$
 72. $(x^2 + y^2 - 4)(x^2 + 9y^2 - 9) \leq 0$
 73. $4y^2 - x^2 \geq 4$ 74. $|x^2 - y^2| \leq 1$

Theory and Examples

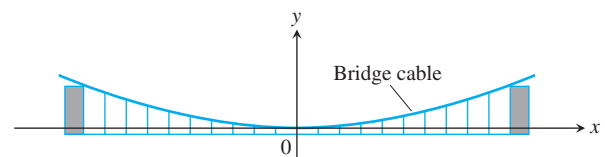
75. **Archimedes' formula for the volume of a parabolic solid** The region enclosed by the parabola $y = (4h/b^2)x^2$ and the line $y = h$ is revolved about the y -axis to generate the solid shown here. Show that the volume of the solid is $3/2$ the volume of the corresponding cone.



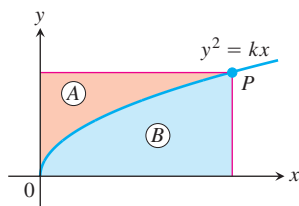
76. **Suspension bridge cables hang in parabolas** The suspension bridge cable shown here supports a uniform load of w pounds per horizontal foot. It can be shown that if H is the horizontal tension of the cable at the origin, then the curve of the cable satisfies the equation

$$\frac{dy}{dx} = \frac{w}{H}x.$$

Show that the cable hangs in a parabola by solving this differential equation subject to the initial condition that $y = 0$ when $x = 0$.

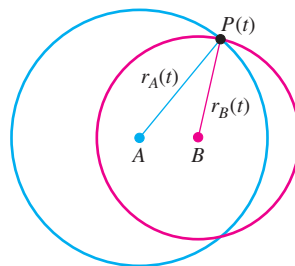


77. Find an equation for the circle through the points $(1, 0)$, $(0, 1)$, and $(2, 2)$.
78. Find an equation for the circle through the points $(2, 3)$, $(3, 2)$, and $(-4, 3)$.
79. Find an equation for the circle centered at $(-2, 1)$ that passes through the point $(1, 3)$. Is the point $(1.1, 2.8)$ inside, outside, or on the circle?
80. Find equations for the tangents to the circle $(x - 2)^2 + (y - 1)^2 = 5$ at the points where the circle crosses the coordinate axes. (*Hint:* Use implicit differentiation.)
81. If lines are drawn parallel to the coordinate axes through a point P on the parabola $y^2 = kx$, $k > 0$, the parabola partitions the rectangular region bounded by these lines and the coordinate axes into two smaller regions, A and B .
- If the two smaller regions are revolved about the y -axis, show that they generate solids whose volumes have the ratio 4:1.
 - What is the ratio of the volumes generated by revolving the regions about the x -axis?



82. Show that the tangents to the curve $y^2 = 4px$ from any point on the line $x = -p$ are perpendicular.
83. Find the dimensions of the rectangle of largest area that can be inscribed in the ellipse $x^2 + 4y^2 = 4$ with its sides parallel to the coordinate axes. What is the area of the rectangle?
84. Find the volume of the solid generated by revolving the region enclosed by the ellipse $9x^2 + 4y^2 = 36$ about the (a) x -axis, (b) y -axis.
85. The “triangular” region in the first quadrant bounded by the x -axis, the line $x = 4$, and the hyperbola $9x^2 - 4y^2 = 36$ is revolved about the x -axis to generate a solid. Find the volume of the solid.
86. The region bounded on the left by the y -axis, on the right by the hyperbola $x^2 - y^2 = 1$, and above and below by the lines $y = \pm 3$ is revolved about the y -axis to generate a solid. Find the volume of the solid.
87. Find the centroid of the region that is bounded below by the x -axis and above by the ellipse $(x^2/9) + (y^2/16) = 1$.
88. The curve $y = \sqrt{x^2 + 1}$, $0 \leq x \leq \sqrt{2}$, which is part of the upper branch of the hyperbola $y^2 - x^2 = 1$, is revolved about the x -axis to generate a surface. Find the area of the surface.
89. The circular waves in the photograph here were made by touching the surface of a ripple tank, first at A and then at B . As the waves

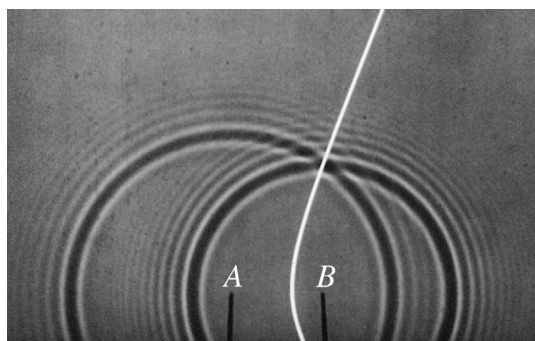
expanded, their point of intersection appeared to trace a hyperbola. Did it really do that? To find out, we can model the waves with circles centered at A and B .



At time t , the point P is $r_A(t)$ units from A and $r_B(t)$ units from B . Since the radii of the circles increase at a constant rate, the rate at which the waves are traveling is

$$\frac{dr_A}{dt} = \frac{dr_B}{dt}.$$

Conclude from this equation that $r_A - r_B$ has a constant value, so that P must lie on a hyperbola with foci at A and B .

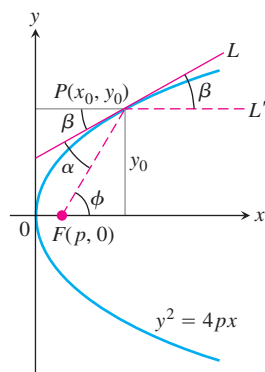


90. **The reflective property of parabolas** The figure here shows a typical point $P(x_0, y_0)$ on the parabola $y^2 = 4px$. The line L is tangent to the parabola at P . The parabola's focus lies at $F(p, 0)$. The ray L' extending from P to the right is parallel to the x -axis. We show that light from F to P will be reflected out along L' by showing that β equals α . Establish this equality by taking the following steps.
- Show that $\tan \beta = 2p/y_0$.
 - Show that $\tan \phi = y_0/(x_0 - p)$.
 - Use the identity

$$\tan \alpha = \frac{\tan \phi - \tan \beta}{1 + \tan \phi \tan \beta}$$

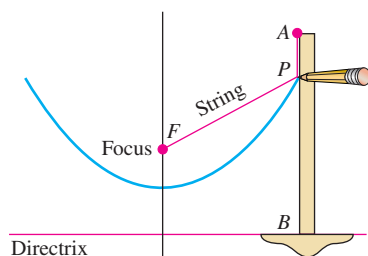
to show that $\tan \alpha = 2p/y_0$.

Since α and β are both acute, $\tan \beta = \tan \alpha$ implies $\beta = \alpha$.

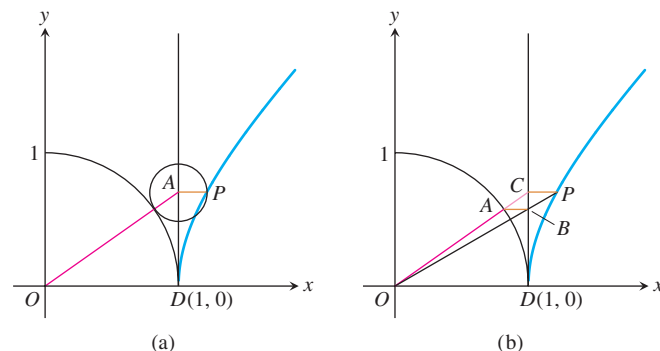


91. How the astronomer Kepler used string to draw parabolas

Kepler's method for drawing a parabola (with more modern tools) requires a string the length of a T square and a table whose edge can serve as the parabola's directrix. Pin one end of the string to the point where you want the focus to be and the other end to the upper end of the T square. Then, holding the string taut against the T square with a pencil, slide the T square along the table's edge. As the T square moves, the pencil will trace a parabola. Why?



92. Construction of a hyperbola The following diagrams appeared (unlabeled) in Ernest J. Eckert, "Constructions Without Words," *Mathematics Magazine*, Vol. 66, No. 2, Apr. 1993, p. 113. Explain the constructions by finding the coordinates of the point P .



93. The width of a parabola at the focus Show that the number $4p$ is the *width* of the parabola $x^2 = 4py$ ($p > 0$) at the focus by showing that the line $y = p$ cuts the parabola at points that are $4p$ units apart.

94. The asymptotes of $(x^2/a^2) - (y^2/b^2) = 1$ Show that the vertical distance between the line $y = (b/a)x$ and the upper half of the right-hand branch $y = (b/a)\sqrt{x^2 - a^2}$ of the hyperbola $(x^2/a^2) - (y^2/b^2) = 1$ approaches 0 by showing that

$$\lim_{x \rightarrow \infty} \left(\frac{b}{a}x - \frac{b}{a}\sqrt{x^2 - a^2} \right) = \frac{b}{a} \lim_{x \rightarrow \infty} (x - \sqrt{x^2 - a^2}) = 0.$$

Similar results hold for the remaining portions of the hyperbola and the lines $y = \pm(b/a)x$.

10.2

Classifying Conic Sections by Eccentricity

We now show how to associate with each conic section a number called the conic section's *eccentricity*. The eccentricity reveals the conic section's type (circle, ellipse, parabola, or hyperbola) and, in the case of ellipses and hyperbolas, describes the conic section's general proportions.

Eccentricity

Although the center-to-focus distance c does not appear in the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad (a > b)$$

for an ellipse, we can still determine c from the equation $c = \sqrt{a^2 - b^2}$. If we fix a and vary c over the interval $0 \leq c \leq a$, the resulting ellipses will vary in shape (Figure 10.17). They are circles if $c = 0$ (so that $a = b$) and flatten as c increases. If $c = a$, the foci and vertices overlap and the ellipse degenerates into a line segment.

We use the ratio of c to a to describe the various shapes the ellipse can take. We call this ratio the ellipse's eccentricity.

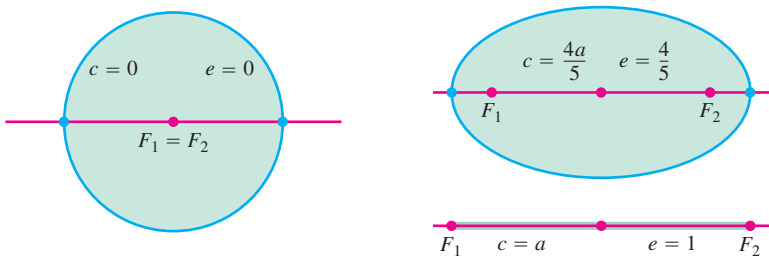


FIGURE 10.17 The ellipse changes from a circle to a line segment as c increases from 0 to a .

DEFINITION Eccentricity of an Ellipse

The **eccentricity** of the ellipse $(x^2/a^2) + (y^2/b^2) = 1$ ($a > b$) is

$$e = \frac{c}{a} = \frac{\sqrt{a^2 - b^2}}{a}.$$

TABLE 10.2 Eccentricities of planetary orbits

Mercury	0.21	Saturn	0.06
Venus	0.01	Uranus	0.05
Earth	0.02	Neptune	0.01
Mars	0.09	Pluto	0.25
Jupiter	0.05		

The planets in the solar system revolve around the sun in (approximate) elliptical orbits with the sun at one focus. Most of the orbits are nearly circular, as can be seen from the eccentricities in Table 10.2. Pluto has a fairly eccentric orbit, with $e = 0.25$, as does Mercury, with $e = 0.21$. Other members of the solar system have orbits that are even more eccentric. Icarus, an asteroid about 1 mile wide that revolves around the sun every 409 Earth days, has an orbital eccentricity of 0.83 (Figure 10.18).

EXAMPLE 1 Halley's Comet

The orbit of Halley's comet is an ellipse 36.18 astronomical units long by 9.12 astronomical units wide. (One *astronomical unit* [AU] is 149,597,870 km, the semimajor axis of Earth's orbit.) Its eccentricity is

$$e = \frac{\sqrt{a^2 - b^2}}{a} = \frac{\sqrt{(36.18/2)^2 - (9.12/2)^2}}{(1/2)(36.18)} = \frac{\sqrt{(18.09)^2 - (4.56)^2}}{18.09} \approx 0.97. \quad \blacksquare$$

Whereas a parabola has one focus and one directrix, each **ellipse** has two foci and two **directrices**. These are the lines perpendicular to the major axis at distances $\pm a/e$ from the center. The parabola has the property that

$$PF = 1 \cdot PD \tag{1}$$

for any point P on it, where F is the focus and D is the point nearest P on the directrix. For an ellipse, it can be shown that the equations that replace Equation (1) are

$$PF_1 = e \cdot PD_1, \quad PF_2 = e \cdot PD_2. \tag{2}$$

Here, e is the eccentricity, P is any point on the ellipse, F_1 and F_2 are the foci, and D_1 and D_2 are the points on the directrices nearest P (Figure 10.19).

In both Equations (2) the directrix and focus must correspond; that is, if we use the distance from P to F_1 , we must also use the distance from P to the directrix at the same end of the ellipse. The directrix $x = -a/e$ corresponds to $F_1(-c, 0)$, and the directrix $x = a/e$ corresponds to $F_2(c, 0)$.

The eccentricity of a hyperbola is also $e = c/a$, only in this case c equals $\sqrt{a^2 + b^2}$ instead of $\sqrt{a^2 - b^2}$. In contrast to the eccentricity of an ellipse, the eccentricity of a hyperbola is always greater than 1.

HISTORICAL BIOGRAPHY

Edmund Halley
(1656–1742)

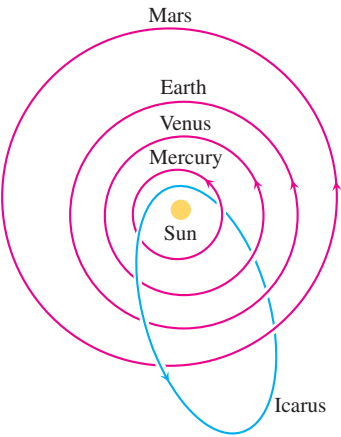


FIGURE 10.18 The orbit of the asteroid Icarus is highly eccentric. Earth's orbit is so nearly circular that its foci lie inside the sun.

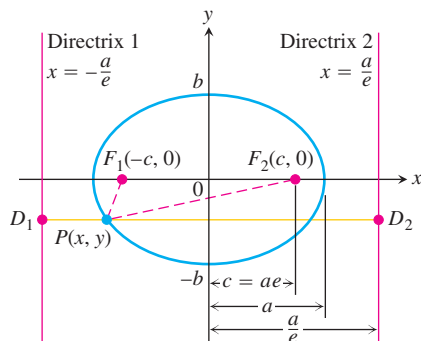


FIGURE 10.19 The foci and directrices of the ellipse $(x^2/a^2) + (y^2/b^2) = 1$. Directrix 1 corresponds to focus F_1 , and directrix 2 to focus F_2 .

DEFINITION Eccentricity of a Hyperbola

The **eccentricity** of the hyperbola $(x^2/a^2) - (y^2/b^2) = 1$ is

$$e = \frac{c}{a} = \frac{\sqrt{a^2 + b^2}}{a}.$$

In both ellipse and hyperbola, the eccentricity is the ratio of the distance between the foci to the distance between the vertices (because $c/a = 2c/2a$).

$$\text{Eccentricity} = \frac{\text{distance between foci}}{\text{distance between vertices}}$$

In an ellipse, the foci are closer together than the vertices and the ratio is less than 1. In a hyperbola, the foci are farther apart than the vertices and the ratio is greater than 1.

EXAMPLE 2 Finding the Vertices of an Ellipse

Locate the vertices of an ellipse of eccentricity 0.8 whose foci lie at the points $(0, \pm 7)$.

Solution Since $e = c/a$, the vertices are the points $(0, \pm a)$ where

$$a = \frac{c}{e} = \frac{7}{0.8} = 8.75,$$

or $(0, \pm 8.75)$. ■

EXAMPLE 3 Eccentricity of a Hyperbola

Find the eccentricity of the hyperbola $9x^2 - 16y^2 = 144$.

Solution We divide both sides of the hyperbola's equation by 144 to put it in standard form, obtaining

$$\frac{9x^2}{144} - \frac{16y^2}{144} = 1 \quad \text{and} \quad \frac{x^2}{16} - \frac{y^2}{9} = 1.$$

With $a^2 = 16$ and $b^2 = 9$, we find that $c = \sqrt{a^2 + b^2} = \sqrt{16 + 9} = 5$, so

$$e = \frac{c}{a} = \frac{5}{4}. \quad \blacksquare$$

As with the ellipse, it can be shown that the lines $x = \pm a/e$ act as **directrices** for the hyperbola and that

$$PF_1 = e \cdot PD_1 \quad \text{and} \quad PF_2 = e \cdot PD_2. \quad (3)$$

Here P is any point on the hyperbola, F_1 and F_2 are the foci, and D_1 and D_2 are the points nearest P on the directrices (Figure 10.20).

To complete the picture, we define the eccentricity of a parabola to be $e = 1$. Equations (1) to (3) then have the common form $PF = e \cdot PD$.

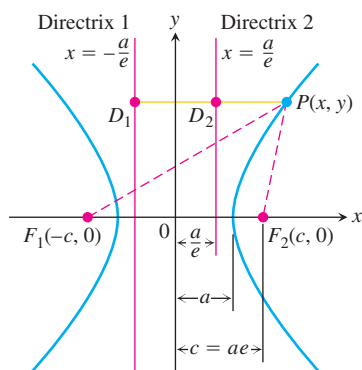


FIGURE 10.20 The foci and directrices of the hyperbola $(x^2/a^2) - (y^2/b^2) = 1$. No matter where P lies on the hyperbola, $PF_1 = e \cdot PD_1$ and $PF_2 = e \cdot PD_2$.

DEFINITION Eccentricity of a Parabola

The **eccentricity** of a parabola is $e = 1$.

The “focus–directrix” equation $PF = e \cdot PD$ unites the parabola, ellipse, and hyperbola in the following way. Suppose that the distance PF of a point P from a fixed point F (the focus) is a constant multiple of its distance from a fixed line (the directrix). That is, suppose

$$PF = e \cdot PD, \quad (4)$$

where e is the constant of proportionality. Then the path traced by P is

- (a) a *parabola* if $e = 1$,
- (b) an *ellipse* of eccentricity e if $e < 1$, and
- (c) a *hyperbola* of eccentricity e if $e > 1$.

There are no coordinates in Equation (4) and when we try to translate it into coordinate form it translates in different ways, depending on the size of e . At least, that is what happens in Cartesian coordinates. However, in polar coordinates, as we will see in Section 10.8, the equation $PF = e \cdot PD$ translates into a single equation regardless of the value of e , an equation so simple that it has been the equation of choice of astronomers and space scientists for nearly 300 years.

Given the focus and corresponding directrix of a hyperbola centered at the origin and with foci on the x -axis, we can use the dimensions shown in Figure 10.20 to find e . Knowing e , we can derive a Cartesian equation for the hyperbola from the equation $PF = e \cdot PD$, as in the next example. We can find equations for ellipses centered at the origin and with foci on the x -axis in a similar way, using the dimensions shown in Figure 10.19.

EXAMPLE 4 Cartesian Equation for a Hyperbola

Find a Cartesian equation for the hyperbola centered at the origin that has a focus at $(3, 0)$ and the line $x = 1$ as the corresponding directrix.

Solution We first use the dimensions shown in Figure 10.20 to find the hyperbola’s eccentricity. The focus is

$$(c, 0) = (3, 0) \quad \text{so} \quad c = 3.$$

The directrix is the line

$$x = \frac{a}{e} = 1, \quad \text{so} \quad a = e.$$

When combined with the equation $e = c/a$ that defines eccentricity, these results give

$$e = \frac{c}{a} = \frac{3}{e}, \quad \text{so} \quad e^2 = 3 \quad \text{and} \quad e = \sqrt{3}.$$

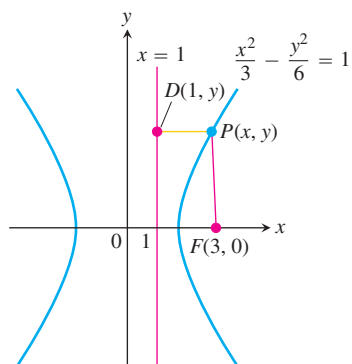


FIGURE 10.21 The hyperbola and directrix in Example 4.

Knowing e , we can now derive the equation we want from the equation $PF = e \cdot PD$. In the notation of Figure 10.21, we have

$$PF = e \cdot PD \quad \text{Equation (4)}$$

$$\sqrt{(x-3)^2 + (y-0)^2} = \sqrt{3} |x-1| \quad e = \sqrt{3}$$

$$x^2 - 6x + 9 + y^2 = 3(x^2 - 2x + 1)$$

$$2x^2 - y^2 = 6$$

$$\frac{x^2}{3} - \frac{y^2}{6} = 1.$$



EXERCISES 10.2

Ellipses

In Exercises 1–8, find the eccentricity of the ellipse. Then find and graph the ellipse's foci and directrices.

- | | |
|--------------------------|----------------------------|
| 1. $16x^2 + 25y^2 = 400$ | 2. $7x^2 + 16y^2 = 112$ |
| 3. $2x^2 + y^2 = 2$ | 4. $2x^2 + y^2 = 4$ |
| 5. $3x^2 + 2y^2 = 6$ | 6. $9x^2 + 10y^2 = 90$ |
| 7. $6x^2 + 9y^2 = 54$ | 8. $169x^2 + 25y^2 = 4225$ |

Exercises 9–12 give the foci or vertices and the eccentricities of ellipses centered at the origin of the xy -plane. In each case, find the ellipse's standard-form equation.

- | | |
|--|---|
| 9. Foci: $(0, \pm 3)$
Eccentricity: 0.5 | 10. Foci: $(\pm 8, 0)$
Eccentricity: 0.2 |
| 11. Vertices: $(0, \pm 70)$
Eccentricity: 0.1 | 12. Vertices: $(\pm 10, 0)$
Eccentricity: 0.24 |

Exercises 13–16 give foci and corresponding directrices of ellipses centered at the origin of the xy -plane. In each case, use the dimensions in Figure 10.19 to find the eccentricity of the ellipse. Then find the ellipse's standard-form equation.

- | | |
|---|--|
| 13. Focus: $(\sqrt{5}, 0)$
Directrix: $x = \frac{9}{\sqrt{5}}$ | 14. Focus: $(4, 0)$
Directrix: $x = \frac{16}{3}$ |
| 15. Focus: $(-4, 0)$
Directrix: $x = -16$ | 16. Focus: $(-\sqrt{2}, 0)$
Directrix: $x = -2\sqrt{2}$ |
17. Draw an ellipse of eccentricity $4/5$. Explain your procedure.
18. Draw the orbit of Pluto (eccentricity 0.25) to scale. Explain your procedure.
19. The endpoints of the major and minor axes of an ellipse are $(1, 1)$, $(3, 4)$, $(1, 7)$, and $(-1, 4)$. Sketch the ellipse, give its equation in standard form, and find its foci, eccentricity, and directrices.

20. Find an equation for the ellipse of eccentricity $2/3$ that has the line $x = 9$ as a directrix and the point $(4, 0)$ as the corresponding focus.

21. What values of the constants a , b , and c make the ellipse

$$4x^2 + y^2 + ax + by + c = 0$$

lie tangent to the x -axis at the origin and pass through the point $(-1, 2)$? What is the eccentricity of the ellipse?

22. **The reflective property of ellipses** An ellipse is revolved about its major axis to generate an ellipsoid. The inner surface of the ellipsoid is silvered to make a mirror. Show that a ray of light emanating from one focus will be reflected to the other focus. Sound waves also follow such paths, and this property is used in constructing “whispering galleries.” (*Hint:* Place the ellipse in standard position in the xy -plane and show that the lines from a point P on the ellipse to the two foci make congruent angles with the tangent to the ellipse at P .)

Hyperbolas

In Exercises 23–30, find the eccentricity of the hyperbola. Then find and graph the hyperbola's foci and directrices.

- | | |
|------------------------|----------------------------|
| 23. $x^2 - y^2 = 1$ | 24. $9x^2 - 16y^2 = 144$ |
| 25. $y^2 - x^2 = 8$ | 26. $y^2 - x^2 = 4$ |
| 27. $8x^2 - 2y^2 = 16$ | 28. $y^2 - 3x^2 = 3$ |
| 29. $8y^2 - 2x^2 = 16$ | 30. $64x^2 - 36y^2 = 2304$ |

Exercises 31–34 give the eccentricities and the vertices or foci of hyperbolas centered at the origin of the xy -plane. In each case, find the hyperbola's standard-form equation.

- | | |
|---|---|
| 31. Eccentricity: 3
Vertices: $(0, \pm 1)$ | 32. Eccentricity: 2
Vertices: $(\pm 2, 0)$ |
| 33. Eccentricity: 3
Foci: $(\pm 3, 0)$ | 34. Eccentricity: 1.25
Foci: $(0, \pm 5)$ |

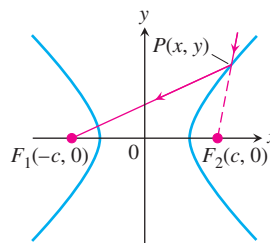
Exercises 35–38 give foci and corresponding directrices of hyperbolas centered at the origin of the xy -plane. In each case, find the hyperbola's eccentricity. Then find the hyperbola's standard-form equation.

35. Focus: $(4, 0)$ 36. Focus: $(\sqrt{10}, 0)$
 Directrix: $x = 2$ Directrix: $x = \sqrt{2}$
 37. Focus: $(-2, 0)$ 38. Focus: $(-6, 0)$
 Directrix: $x = -\frac{1}{2}$ Directrix: $x = -2$

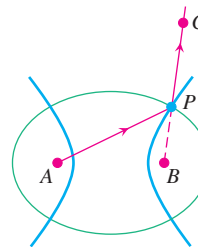
39. A hyperbola of eccentricity $3/2$ has one focus at $(1, -3)$. The corresponding directrix is the line $y = 2$. Find an equation for the hyperbola.

T 40. The effect of eccentricity on a hyperbola's shape What happens to the graph of a hyperbola as its eccentricity increases? To find out, rewrite the equation $(x^2/a^2) - (y^2/b^2) = 1$ in terms of a and e instead of a and b . Graph the hyperbola for various values of e and describe what you find.

41. **The reflective property of hyperbolas** Show that a ray of light directed toward one focus of a hyperbolic mirror, as in the accompanying figure, is reflected toward the other focus. (*Hint:* Show that the tangent to the hyperbola at P bisects the angle made by segments PF_1 and PF_2 .)



42. **A confocal ellipse and hyperbola** Show that an ellipse and a hyperbola that have the same foci A and B , as in the accompanying figure, cross at right angles at their point of intersection. (*Hint:* A ray of light from focus A that met the hyperbola at P would be reflected from the hyperbola as if it came directly from B (Exercise 41). The same ray would be reflected off the ellipse to pass through B (Exercise 22).)



10.3 Quadratic Equations and Rotations

In this section, we examine the Cartesian graph of any equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0, \quad (1)$$

in which A , B , and C are not all zero, and show that it is nearly always a conic section. The exceptions are the cases in which there is no graph at all or the graph consists of two parallel lines. It is conventional to call all graphs of Equation (1), curved or not, **quadratic curves**.

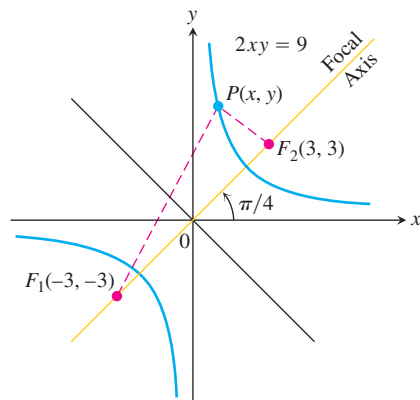


FIGURE 10.22 The focal axis of the hyperbola $2xy = 9$ makes an angle of $\pi/4$ radians with the positive x -axis.

The Cross Product Term

You may have noticed that the term Bxy did not appear in the equations for the conic sections in Section 10.1. This happened because the axes of the conic sections ran parallel to (in fact, coincided with) the coordinate axes.

To see what happens when the parallelism is absent, let us write an equation for a hyperbola with $a = 3$ and foci at $F_1(-3, -3)$ and $F_2(3, 3)$ (Figure 10.22). The equation $|PF_1 - PF_2| = 2a$ becomes $|PF_1 - PF_2| = 2(3) = 6$ and

$$\sqrt{(x+3)^2 + (y+3)^2} - \sqrt{(x-3)^2 + (y-3)^2} = \pm 6.$$

When we transpose one radical, square, solve for the radical that still appears, and square again, the equation reduces to

$$2xy = 9, \quad (2)$$

a case of Equation (1) in which the cross product term is present. The asymptotes of the hyperbola in Equation (2) are the x - and y -axes, and the focal axis makes an angle of $\pi/4$

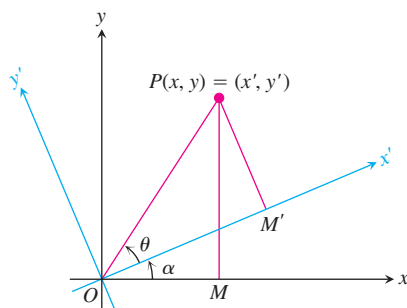


FIGURE 10.23 A counterclockwise rotation through angle α about the origin.

radians with the positive x -axis. As in this example, the cross product term is present in Equation (1) only when the axes of the conic are tilted.

To eliminate the xy -term from the equation of a conic, we rotate the coordinate axes to eliminate the “tilt” in the axes of the conic. The equations for the rotations we use are derived in the following way. In the notation of Figure 10.23, which shows a counterclockwise rotation about the origin through an angle α ,

$$\begin{aligned} x &= OM = OP \cos(\theta + \alpha) = OP \cos \theta \cos \alpha - OP \sin \theta \sin \alpha \\ y &= MP = OP \sin(\theta + \alpha) = OP \cos \theta \sin \alpha + OP \sin \theta \cos \alpha. \end{aligned} \quad (3)$$

Since

$$OP \cos \theta = OM' = x'$$

and

$$OP \sin \theta = M'P = y',$$

Equations (3) reduce to the following.

Equations for Rotating Coordinate Axes

$$\begin{aligned} x &= x' \cos \alpha - y' \sin \alpha \\ y &= x' \sin \alpha + y' \cos \alpha \end{aligned} \quad (4)$$

EXAMPLE 1 Finding an Equation for a Hyperbola

The x - and y -axes are rotated through an angle of $\pi/4$ radians about the origin. Find an equation for the hyperbola $2xy = 9$ in the new coordinates.

Solution Since $\cos \pi/4 = \sin \pi/4 = 1/\sqrt{2}$, we substitute

$$x = \frac{x' - y'}{\sqrt{2}}, \quad y = \frac{x' + y'}{\sqrt{2}}$$

from Equations (4) into the equation $2xy = 9$ and obtain

$$\begin{aligned} 2 \left(\frac{x' - y'}{\sqrt{2}} \right) \left(\frac{x' + y'}{\sqrt{2}} \right) &= 9 \\ x'^2 - y'^2 &= 9 \\ \frac{x'^2}{9} - \frac{y'^2}{9} &= 1. \end{aligned}$$

See Figure 10.24.

If we apply Equations (4) to the quadratic equation (1), we obtain a new quadratic equation

$$A'x'^2 + B'x'y' + C'y'^2 + D'x' + E'y' + F' = 0. \quad (5)$$

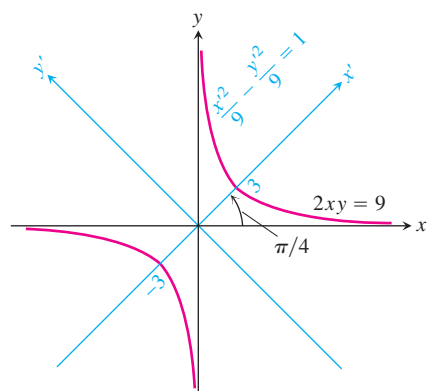


FIGURE 10.24 The hyperbola in Example 1 (x' and y' are the coordinates).

The new and old coefficients are related by the equations

$$\begin{aligned}
 A' &= A \cos^2 \alpha + B \cos \alpha \sin \alpha + C \sin^2 \alpha \\
 B' &= B \cos 2\alpha + (C - A) \sin 2\alpha \\
 C' &= A \sin^2 \alpha - B \sin \alpha \cos \alpha + C \cos^2 \alpha \\
 D' &= D \cos \alpha + E \sin \alpha \\
 E' &= -D \sin \alpha + E \cos \alpha \\
 F' &= F.
 \end{aligned} \tag{6}$$

These equations show, among other things, that if we start with an equation for a curve in which the cross product term is present ($B \neq 0$), we can find a rotation angle α that produces an equation in which no cross product term appears ($B' = 0$). To find α , we set $B' = 0$ in the second equation in (6) and solve the resulting equation,

$$B \cos 2\alpha + (C - A) \sin 2\alpha = 0,$$

for α . In practice, this means determining α from one of the two equations

Angle of Rotation

$$\cot 2\alpha = \frac{A - C}{B} \quad \text{or} \quad \tan 2\alpha = \frac{B}{A - C}. \tag{7}$$

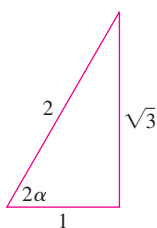


FIGURE 10.25 This triangle identifies $2\alpha = \cot^{-1}(1/\sqrt{3})$ as $\pi/3$ (Example 2).

EXAMPLE 2 Finding the Angle of Rotation

The coordinate axes are to be rotated through an angle α to produce an equation for the curve

$$2x^2 + \sqrt{3}xy + y^2 - 10 = 0$$

that has no cross product term. Find α and the new equation. Identify the curve.

Solution The equation $2x^2 + \sqrt{3}xy + y^2 - 10 = 0$ has $A = 2$, $B = \sqrt{3}$, and $C = 1$. We substitute these values into Equation (7) to find α :

$$\cot 2\alpha = \frac{A - C}{B} = \frac{2 - 1}{\sqrt{3}} = \frac{1}{\sqrt{3}}.$$

From the right triangle in Figure 10.25, we see that one appropriate choice of angle is $2\alpha = \pi/3$, so we take $\alpha = \pi/6$. Substituting $\alpha = \pi/6$, $A = 2$, $B = \sqrt{3}$, $C = 1$, $D = E = 0$, and $F = -10$ into Equations (6) gives

$$A' = \frac{5}{2}, \quad B' = 0, \quad C' = \frac{1}{2}, \quad D' = E' = 0, \quad F' = -10.$$

Equation (5) then gives

$$\frac{5}{2}x'^2 + \frac{1}{2}y'^2 - 10 = 0, \quad \text{or} \quad \frac{x'^2}{4} + \frac{y'^2}{20} = 1.$$

The curve is an ellipse with foci on the new y' -axis (Figure 10.26). ■

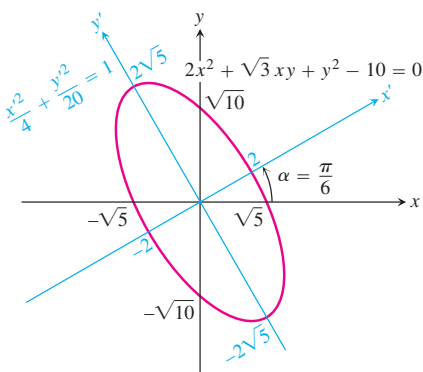


FIGURE 10.26 The conic section in Example 2.

Possible Graphs of Quadratic Equations

We now return to the graph of the general quadratic equation.

Since axes can always be rotated to eliminate the cross product term, there is no loss of generality in assuming that this has been done and that our equation has the form

$$Ax^2 + Cy^2 + Dx + Ey + F = 0. \quad (8)$$

Equation (8) represents

- (a) a *circle* if $A = C \neq 0$ (special cases: the graph is a point or there is no graph at all);
- (b) a *parabola* if Equation (8) is quadratic in one variable and linear in the other;
- (c) an *ellipse* if A and C are both positive or both negative (special cases: circles, a single point, or no graph at all);
- (d) a *hyperbola* if A and C have opposite signs (special case: a pair of intersecting lines);
- (e) a *straight line* if A and C are zero and at least one of D and E is different from zero;
- (f) *one or two straight lines* if the left-hand side of Equation (8) can be factored into the product of two linear factors.

See Table 10.3 for examples.

TABLE 10.3 Examples of quadratic curves $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$

	A	B	C	D	E	F	Equation	Remarks
Circle	1		1			-4	$x^2 + y^2 = 4$	$A = C; F < 0$
Parabola			1	-9			$y^2 = 9x$	Quadratic in y , linear in x
Ellipse	4		9			-36	$4x^2 + 9y^2 = 36$	A, C have same sign, $A \neq C; F < 0$
Hyperbola	1		-1			-1	$x^2 - y^2 = 1$	A, C have opposite signs
One line (still a conic section)	1						$x^2 = 0$	y -axis
Intersecting lines (still a conic section)		1		1	-1	-1	$xy + x - y - 1 = 0$	Factors to $(x - 1)(y + 1) = 0$, so $x = 1, y = -1$
Parallel lines (not a conic section)	1			-3		2	$x^2 - 3x + 2 = 0$	Factors to $(x - 1)(x - 2) = 0$, so $x = 1, x = 2$
Point	1		1				$x^2 + y^2 = 0$	The origin
No graph	1					1	$x^2 = -1$	No graph

The Discriminant Test

We do not need to eliminate the xy -term from the equation

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0 \quad (9)$$

to tell what kind of conic section the equation represents. If this is the only information we want, we can apply the following test instead.

As we have seen, if $B \neq 0$, then rotating the coordinate axes through an angle α that satisfies the equation

$$\cot 2\alpha = \frac{A - C}{B} \quad (10)$$

will change Equation (9) into an equivalent form

$$A'x'^2 + C'y'^2 + D'x' + E'y' + F' = 0 \quad (11)$$

without a cross product term.

Now, the graph of Equation (11) is a (real or degenerate)

- (a) *parabola* if A' or $C' = 0$; that is, if $A'C' = 0$;
- (b) *ellipse* if A' and C' have the same sign; that is, if $A'C' > 0$;
- (c) *hyperbola* if A' and C' have opposite signs; that is, if $A'C' < 0$.

It can also be verified from Equations (6) that for any rotation of axes,

$$B^2 - 4AC = B'^2 - 4A'C'. \quad (12)$$

This means that the quantity $B^2 - 4AC$ is not changed by a rotation. But when we rotate through the angle α given by Equation (10), B' becomes zero, so

$$B^2 - 4AC = -4A'C'.$$

Since the curve is a parabola if $A'C' = 0$, an ellipse if $A'C' > 0$, and a hyperbola if $A'C' < 0$, the curve must be a parabola if $B^2 - 4AC = 0$, an ellipse if $B^2 - 4AC < 0$, and a hyperbola if $B^2 - 4AC > 0$. The number $B^2 - 4AC$ is called the **discriminant** of Equation (9).

The Discriminant Test

With the understanding that occasional degenerate cases may arise, the quadratic curve $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ is

- (a) a **parabola** if $B^2 - 4AC = 0$,
- (b) an **ellipse** if $B^2 - 4AC < 0$,
- (c) a **hyperbola** if $B^2 - 4AC > 0$.

EXAMPLE 3 Applying the Discriminant Test

- (a) $3x^2 - 6xy + 3y^2 + 2x - 7 = 0$ represents a parabola because

$$B^2 - 4AC = (-6)^2 - 4 \cdot 3 \cdot 3 = 36 - 36 = 0.$$

- (b) $x^2 + xy + y^2 - 1 = 0$ represents an ellipse because

$$B^2 - 4AC = (1)^2 - 4 \cdot 1 \cdot 1 = -3 < 0.$$

- (c) $xy - y^2 - 5y + 1 = 0$ represents a hyperbola because

$$B^2 - 4AC = (1)^2 - 4(0)(-1) = 1 > 0. \quad \blacksquare$$

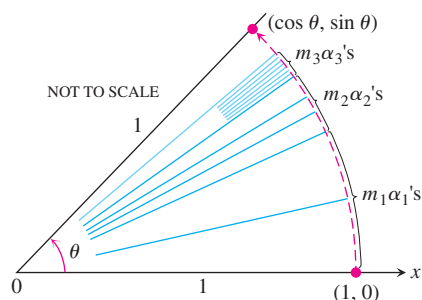


FIGURE 10.27 To calculate the sine and cosine of an angle θ between 0 and 2π , the calculator rotates the point $(1, 0)$ to an appropriate location on the unit circle and displays the resulting coordinates.

USING TECHNOLOGY How Calculators Use Rotations to Evaluate Sines and Cosines

Some calculators use rotations to calculate sines and cosines of arbitrary angles. The procedure goes something like this: The calculator has, stored,

1. ten angles or so, say

$$\alpha_1 = \sin^{-1}(10^{-1}), \quad \alpha_2 = \sin^{-1}(10^{-2}), \quad \dots, \quad \alpha_{10} = \sin^{-1}(10^{-10}),$$

and

2. twenty numbers, the sines and cosines of the angles $\alpha_1, \alpha_2, \dots, \alpha_{10}$.

To calculate the sine and cosine of an arbitrary angle θ , we enter θ (in radians) into the calculator. The calculator subtracts or adds multiples of 2π to θ to replace θ by the angle between 0 and 2π that has the same sine and cosine as θ (we continue to call the angle θ). The calculator then “writes” θ as a sum of multiples of α_1 (as many as possible without overshooting) plus multiples of α_2 (again, as many as possible), and so on, working its way to α_{10} . This gives

$$\theta \approx m_1\alpha_1 + m_2\alpha_2 + \dots + m_{10}\alpha_{10}.$$

The calculator then rotates the point $(1, 0)$ through m_1 copies of α_1 (through α_1 , m_1 times in succession), plus m_2 copies of α_2 , and so on, finishing off with m_{10} copies of α_{10} (Figure 10.27). The coordinates of the final position of $(1, 0)$ on the unit circle are the values the calculator gives for $(\cos \theta, \sin \theta)$.

EXERCISES 10.3

Using the Discriminant

Use the discriminant $B^2 - 4AC$ to decide whether the equations in Exercises 1–16 represent parabolas, ellipses, or hyperbolas.

1. $x^2 - 3xy + y^2 - x = 0$
2. $3x^2 - 18xy + 27y^2 - 5x + 7y = -4$
3. $3x^2 - 7xy + \sqrt{17}y^2 = 1$
4. $2x^2 - \sqrt{15}xy + 2y^2 + x + y = 0$
5. $x^2 + 2xy + y^2 + 2x - y + 2 = 0$
6. $2x^2 - y^2 + 4xy - 2x + 3y = 6$
7. $x^2 + 4xy + 4y^2 - 3x = 6$
8. $x^2 + y^2 + 3x - 2y = 10$
9. $xy + y^2 - 3x = 5$
10. $3x^2 + 6xy + 3y^2 - 4x + 5y = 12$
11. $3x^2 - 5xy + 2y^2 - 7x - 14y = -1$
12. $2x^2 - 4.9xy + 3y^2 - 4x = 7$
13. $x^2 - 3xy + 3y^2 + 6y = 7$
14. $25x^2 + 21xy + 4y^2 - 350x = 0$
15. $6x^2 + 3xy + 2y^2 + 17y + 2 = 0$
16. $3x^2 + 12xy + 12y^2 + 435x - 9y + 72 = 0$

Rotating Coordinate Axes

In Exercises 17–26, rotate the coordinate axes to change the given equation into an equation that has no cross product (xy) term. Then identify the graph of the equation. (The new equations will vary with the size and direction of the rotation you use.)

17. $xy = 2$
18. $x^2 + xy + y^2 = 1$
19. $3x^2 + 2\sqrt{3}xy + y^2 - 8x + 8\sqrt{3}y = 0$
20. $x^2 - \sqrt{3}xy + 2y^2 = 1$
21. $x^2 - 2xy + y^2 = 2$
22. $3x^2 - 2\sqrt{3}xy + y^2 = 1$
23. $\sqrt{2}x^2 + 2\sqrt{2}xy + \sqrt{2}y^2 - 8x + 8y = 0$
24. $xy - y - x + 1 = 0$
25. $3x^2 + 2xy + 3y^2 = 19$
26. $3x^2 + 4\sqrt{3}xy - y^2 = 7$
27. Find the sine and cosine of an angle in Quadrant I through which the coordinate axes can be rotated to eliminate the cross product term from the equation

$$14x^2 + 16xy + 2y^2 - 10x + 26,370y - 17 = 0.$$

Do not carry out the rotation.

28. Find the sine and cosine of an angle in Quadrant II through which the coordinate axes can be rotated to eliminate the cross product term from the equation

$$4x^2 - 4xy + y^2 - 8\sqrt{5}x - 16\sqrt{5}y = 0.$$

Do not carry out the rotation.

T The conic sections in Exercises 17–26 were chosen to have rotation angles that were “nice” in the sense that once we knew $\cot 2\alpha$ or $\tan 2\alpha$ we could identify 2α and find $\sin \alpha$ and $\cos \alpha$ from familiar triangles.

In Exercises 29–34, use a calculator to find an angle α through which the coordinate axes can be rotated to change the given equation into a quadratic equation that has no cross product term. Then find $\sin \alpha$ and $\cos \alpha$ to two decimal places and use Equations (6) to find the coefficients of the new equation to the nearest decimal place. In each case, say whether the conic section is an ellipse, a hyperbola, or a parabola.

29. $x^2 - xy + 3y^2 + x - y - 3 = 0$
 30. $2x^2 + xy - 3y^2 + 3x - 7 = 0$
 31. $x^2 - 4xy + 4y^2 - 5 = 0$
 32. $2x^2 - 12xy + 18y^2 - 49 = 0$
 33. $3x^2 + 5xy + 2y^2 - 8y - 1 = 0$
 34. $2x^2 + 7xy + 9y^2 + 20x - 86 = 0$

Theory and Examples

35. What effect does a 90° rotation about the origin have on the equations of the following conic sections? Give the new equation in each case.

- a. The ellipse $(x^2/a^2) + (y^2/b^2) = 1$ ($a > b$)
 b. The hyperbola $(x^2/a^2) - (y^2/b^2) = 1$
 c. The circle $x^2 + y^2 = a^2$
 d. The line $y = mx$ e. The line $y = mx + b$

36. What effect does a 180° rotation about the origin have on the equations of the following conic sections? Give the new equation in each case.

- a. The ellipse $(x^2/a^2) + (y^2/b^2) = 1$ ($a > b$)
 b. The hyperbola $(x^2/a^2) - (y^2/b^2) = 1$
 c. The circle $x^2 + y^2 = a^2$
 d. The line $y = mx$ e. The line $y = mx + b$

37. **The Hyperbola $xy = a$** The hyperbola $xy = 1$ is one of many hyperbolas of the form $xy = a$ that appear in science and mathematics.

- a. Rotate the coordinate axes through an angle of 45° to change the equation $xy = 1$ into an equation with no xy -term. What is the new equation?
 b. Do the same for the equation $xy = a$.

38. Find the eccentricity of the hyperbola $xy = 2$.

39. Can anything be said about the graph of the equation $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ if $AC < 0$? Give reasons for your answer.

40. **Degenerate conics** Does any nondegenerate conic section $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ have all of the following properties?

- a. It is symmetric with respect to the origin.
 b. It passes through the point $(1, 0)$.
 c. It is tangent to the line $y = 1$ at the point $(-2, 1)$.
 Give reasons for your answer.

41. Show that the equation $x^2 + y^2 = a^2$ becomes $x'^2 + y'^2 = a^2$ for every choice of the angle α in the rotation equations (4).

42. Show that rotating the axes through an angle of $\pi/4$ radians will eliminate the xy -term from Equation (1) whenever $A = C$.

43. a. Decide whether the equation

$$x^2 + 4xy + 4y^2 + 6x + 12y + 9 = 0$$

represents an ellipse, a parabola, or a hyperbola.

- b. Show that the graph of the equation in part (a) is the line $2y = -x - 3$.

44. a. Decide whether the conic section with equation

$$9x^2 + 6xy + y^2 - 12x - 4y + 4 = 0$$

represents a parabola, an ellipse, or a hyperbola.

- b. Show that the graph of the equation in part (a) is the line $y = -3x + 2$.

45. a. What kind of conic section is the curve $xy + 2x - y = 0$?

- b. Solve the equation $xy + 2x - y = 0$ for y and sketch the curve as the graph of a rational function of x .
 c. Find equations for the lines parallel to the line $y = -2x$ that are normal to the curve. Add the lines to your sketch.

46. Prove or find counterexamples to the following statements about the graph of $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$.

- a. If $AC > 0$, the graph is an ellipse.
 b. If $AC > 0$, the graph is a hyperbola.
 c. If $AC < 0$, the graph is a hyperbola.

47. **A nice area formula for ellipses** When $B^2 - 4AC$ is negative, the equation

$$Ax^2 + Bxy + Cy^2 = 1$$

represents an ellipse. If the ellipse's semi-axes are a and b , its area is πab (a standard formula). Show that the area is also given by the formula $2\pi/\sqrt{4AC - B^2}$. (Hint: Rotate the coordinate axes to eliminate the xy -term and apply Equation (12) to the new equation.)

48. **Other invariants** We describe the fact that $B'^2 - 4A'C'$ equals $B^2 - 4AC$ after a rotation about the origin by saying that the discriminant of a quadratic equation is an *invariant* of the equation.

Use Equations (6) to show that the numbers **(a)** $A + C$ and **(b)** $D^2 + E^2$ are also invariants, in the sense that

$$A' + C' = A + C \quad \text{and} \quad D'^2 + E'^2 = D^2 + E^2.$$

We can use these equalities to check against numerical errors when we rotate axes.

49. A proof that $B'^2 - 4A'C' = B^2 - 4AC$ Use Equations (6) to show that $B'^2 - 4A'C' = B^2 - 4AC$ for any rotation of axes about the origin.

10.4

Conics and Parametric Equations; The Cycloid

Curves in the Cartesian plane defined by parametric equations, and the calculation of their derivatives, were introduced in Section 3.5. There we studied parametrizations of lines, circles, and ellipses. In this section we discuss parametrization of parabolas, hyperbolas, cycloids, brachistocrones, and tautochrone.

Parabolas and Hyperbolas

In Section 3.5 we used the parametrization

$$x = \sqrt{t}, \quad y = t, \quad t > 0$$

to describe the motion of a particle moving along the right branch of the parabola $y = x^2$. In the following example we obtain a parametrization of the entire parabola, not just its right branch.

EXAMPLE 1 An Entire Parabola

The position $P(x, y)$ of a particle moving in the xy -plane is given by the equations and parameter interval

$$x = t, \quad y = t^2, \quad -\infty < t < \infty.$$

Identify the particle's path and describe the motion.

Solution We identify the path by eliminating t between the equations $x = t$ and $y = t^2$, obtaining

$$y = (t)^2 = x^2.$$

The particle's position coordinates satisfy the equation $y = x^2$, so the particle moves along this curve.

In contrast to Example 10 in Section 3.5, the particle now traverses the entire parabola. As t increases from $-\infty$ to ∞ , the particle comes down the left-hand side, passes through the origin, and moves up the right-hand side (Figure 10.28). ■

As Example 1 illustrates, any curve $y = f(x)$ has the parametrization $x = t$, $y = f(t)$. This is so simple we usually do not use it, but the point of view is occasionally helpful.

EXAMPLE 2 A Parametrization of the Right-hand Branch of the Hyperbola $x^2 - y^2 = 1$

Describe the motion of the particle whose position $P(x, y)$ at time t is given by

$$x = \sec t, \quad y = \tan t, \quad -\frac{\pi}{2} < t < \frac{\pi}{2}.$$

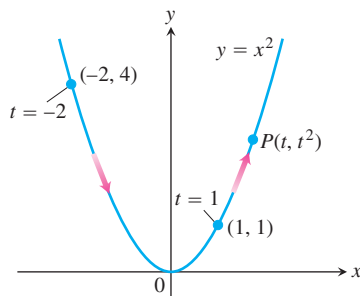


FIGURE 10.28 The path defined by $x = t$, $y = t^2$, $-\infty < t < \infty$ is the entire parabola $y = x^2$ (Example 1).

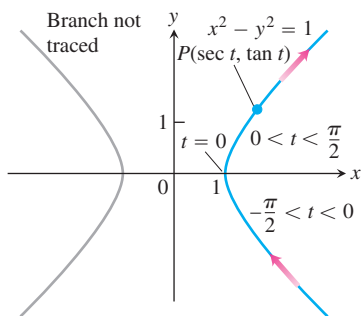


FIGURE 10.29 The equations $x = \sec t$, $y = \tan t$ and interval $-\pi/2 < t < \pi/2$ describe the right-hand branch of the hyperbola $x^2 - y^2 = 1$ (Example 2).

HISTORICAL BIOGRAPHY

Christiaan Huygens
(1629–1695)

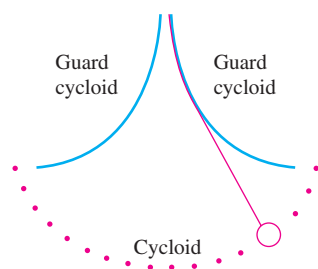


FIGURE 10.30 In Huygens' pendulum clock, the bob swings in a cycloid, so the frequency is independent of the amplitude.

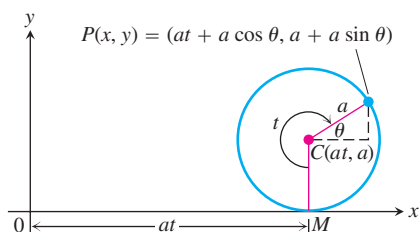


FIGURE 10.31 The position of $P(x, y)$ on the rolling wheel at angle t (Example 3).

Solution We find a Cartesian equation for the coordinates of P by eliminating t between the equations

$$\sec t = x, \quad \tan t = y.$$

We accomplish this with the identity $\sec^2 t - \tan^2 t = 1$, which yields

$$x^2 - y^2 = 1.$$

Since the particle's coordinates (x, y) satisfy the equation $x^2 - y^2 = 1$, the motion takes place somewhere on this hyperbola. As t runs between $-\pi/2$ and $\pi/2$, $x = \sec t$ remains positive and $y = \tan t$ runs between $-\infty$ and ∞ , so P traverses the hyperbola's right-hand branch. It comes in along the branch's lower half as $t \rightarrow 0^-$, reaches $(1, 0)$ at $t = 0$, and moves out into the first quadrant as t increases toward $\pi/2$ (Figure 10.29). ■

Cycloids

The problem with a pendulum clock whose bob swings in a circular arc is that the frequency of the swing depends on the amplitude of the swing. The wider the swing, the longer it takes the bob to return to center (its lowest position).

This does not happen if the bob can be made to swing in a *cycloid*. In 1673, Christiaan Huygens designed a pendulum clock whose bob would swing in a cycloid, a curve we define in Example 3. He hung the bob from a fine wire constrained by guards that caused it to draw up as it swung away from center (Figure 10.30).

EXAMPLE 3 Parametrizing a Cycloid

A wheel of radius a rolls along a horizontal straight line. Find parametric equations for the path traced by a point P on the wheel's circumference. The path is called a **cycloid**.

Solution We take the line to be the x -axis, mark a point P on the wheel, start the wheel with P at the origin, and roll the wheel to the right. As parameter, we use the angle t through which the wheel turns, measured in radians. Figure 10.31 shows the wheel a short while later, when its base lies at units from the origin. The wheel's center C lies at (at, a) and the coordinates of P are

$$x = at + a \cos \theta, \quad y = a + a \sin \theta.$$

To express θ in terms of t , we observe that $t + \theta = 3\pi/2$ in the figure, so that

$$\theta = \frac{3\pi}{2} - t.$$

This makes

$$\cos \theta = \cos \left(\frac{3\pi}{2} - t \right) = -\sin t, \quad \sin \theta = \sin \left(\frac{3\pi}{2} - t \right) = -\cos t.$$

The equations we seek are

$$x = at - a \sin t, \quad y = a - a \cos t.$$

These are usually written with the a factored out:

$$x = a(t - \sin t), \quad y = a(1 - \cos t). \quad (1)$$

Figure 10.32 shows the first arch of the cycloid and part of the next. ■

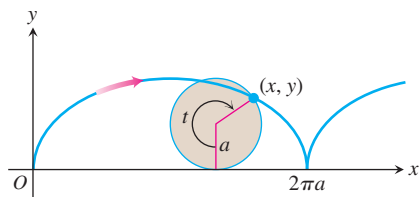


FIGURE 10.32 The cycloid
 $x = a(t - \sin t)$, $y = a(1 - \cos t)$, for
 $t \geq 0$.

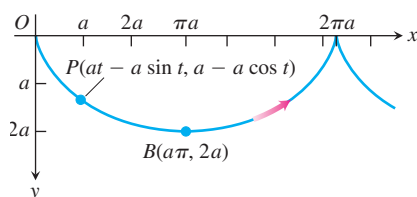


FIGURE 10.33 To study motion along an upside-down cycloid under the influence of gravity, we turn Figure 10.32 upside down. This points the y -axis in the direction of the gravitational force and makes the downward y -coordinates positive. The equations and parameter interval for the cycloid are still

$$\begin{aligned} x &= a(t - \sin t), \\ y &= a(1 - \cos t), \quad t \geq 0. \end{aligned}$$

The arrow shows the direction of increasing t .

Brachistochrones and Tautochrones

If we turn Figure 10.32 upside down, Equations (1) still apply and the resulting curve (Figure 10.33) has two interesting physical properties. The first relates to the origin O and the point B at the bottom of the first arch. Among all smooth curves joining these points, the cycloid is the curve along which a frictionless bead, subject only to the force of gravity, will slide from O to B the fastest. This makes the cycloid a **brachistochrone** (“brah-kiss-toe-krone”), or shortest time curve for these points. The second property is that even if you start the bead partway down the curve toward B , it will still take the bead the same amount of time to reach B . This makes the cycloid a **tautochrone** (“taw-toe-krone”), or same-time curve for O and B .

Are there any other brachistochrones joining O and B , or is the cycloid the only one? We can formulate this as a mathematical question in the following way. At the start, the kinetic energy of the bead is zero, since its velocity is zero. The work done by gravity in moving the bead from $(0, 0)$ to any other point (x, y) in the plane is mgy , and this must equal the change in kinetic energy. That is,

$$mgy = \frac{1}{2}mv^2 - \frac{1}{2}m(0)^2.$$

Thus, the velocity of the bead when it reaches (x, y) has to be

$$v = \sqrt{2gy}.$$

That is,

$$\frac{ds}{dt} = \sqrt{2gy} \quad \text{\textit{ds is the arc length differential along the bead's path.}}$$

or

$$dt = \frac{ds}{\sqrt{2gy}} = \frac{\sqrt{1 + (dy/dx)^2} dx}{\sqrt{2gy}}.$$

The time T_f it takes the bead to slide along a particular path $y = f(x)$ from O to $B(a\pi, 2a)$ is

$$T_f = \int_{x=0}^{x=a\pi} \sqrt{\frac{1 + (dy/dx)^2}{2gy}} dx. \quad (2)$$

What curves $y = f(x)$, if any, minimize the value of this integral?

At first sight, we might guess that the straight line joining O and B would give the shortest time, but perhaps not. There might be some advantage in having the bead fall vertically at first to build up its velocity faster. With a higher velocity, the bead could travel a longer path and still reach B first. Indeed, this is the right idea. The solution, from a branch of mathematics known as the *calculus of variations*, is that the original cycloid from O to B is the one and only brachistochrone for O and B .

While the solution of the brachistochrone problem is beyond our present reach, we can still show why the cycloid is a tautochrone. For the cycloid, Equation (2) takes the form

$$\begin{aligned} T_{\text{cycloid}} &= \int_{x=0}^{x=a\pi} \sqrt{\frac{dx^2 + dy^2}{2gy}} \\ &= \int_{t=0}^{t=\pi} \sqrt{\frac{a^2(2 - 2\cos t)}{2ga(1 - \cos t)}} dt \\ &= \int_0^\pi \sqrt{\frac{a}{g}} dt = \pi\sqrt{\frac{a}{g}}. \end{aligned}$$

From Equations (1),
 $dx = a(1 - \cos t) dt$,
 $dy = a \sin t dt$, and
 $y = a(1 - \cos t)$

Thus, the amount of time it takes the frictionless bead to slide down the cycloid to B after it is released from rest at O is $\pi\sqrt{a/g}$.

Suppose that instead of starting the bead at O we start it at some lower point on the cycloid, a point (x_0, y_0) corresponding to the parameter value $t_0 > 0$. The bead's velocity at any later point (x, y) on the cycloid is

$$v = \sqrt{2g(y - y_0)} = \sqrt{2ga(\cos t_0 - \cos t)}. \quad y = a(1 - \cos t)$$

Accordingly, the time required for the bead to slide from (x_0, y_0) down to B is

$$\begin{aligned} T &= \int_{t_0}^{\pi} \sqrt{\frac{a^2(2 - 2\cos t)}{2ga(\cos t_0 - \cos t)}} dt = \sqrt{\frac{a}{g}} \int_{t_0}^{\pi} \sqrt{\frac{1 - \cos t}{\cos t_0 - \cos t}} dt \\ &= \sqrt{\frac{a}{g}} \int_{t_0}^{\pi} \sqrt{\frac{2\sin^2(t/2)}{(2\cos^2(t_0/2) - 1) - (2\cos^2(t/2) - 1)}} dt \\ &= \sqrt{\frac{a}{g}} \int_{t_0}^{\pi} \frac{\sin(t/2) dt}{\sqrt{\cos^2(t_0/2) - \cos^2(t/2)}} \\ &= \sqrt{\frac{a}{g}} \int_{t_0}^{\pi} \frac{-2 du}{\sqrt{a^2 - u^2}} \quad \begin{array}{l} u = \cos(t/2) \\ -2 du = \sin(t/2) dt \\ c = \cos(t_0/2) \end{array} \\ &= 2\sqrt{\frac{a}{g}} \left[-\sin^{-1} \frac{u}{c} \right]_{t_0}^{\pi} \\ &= 2\sqrt{\frac{a}{g}} \left[-\sin^{-1} \frac{\cos(t/2)}{\cos(t_0/2)} \right]_{t_0}^{\pi} \\ &= 2\sqrt{\frac{a}{g}} (-\sin^{-1} 0 + \sin^{-1} 1) = \pi\sqrt{\frac{a}{g}}. \end{aligned}$$

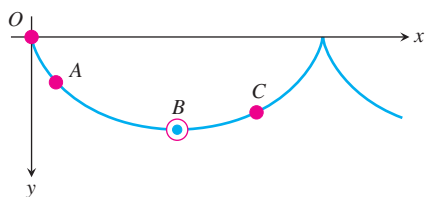


FIGURE 10.34 Beads released simultaneously on the cycloid at O , A , and C will reach B at the same time.

This is precisely the time it takes the bead to slide to B from O . It takes the bead the same amount of time to reach B no matter where it starts. Beads starting simultaneously from O , A , and C in Figure 10.34, for instance, will all reach B at the same time. This is the reason that Huygens' pendulum clock is independent of the amplitude of the swing.

EXERCISES 10.4

Parametric Equations for Conics

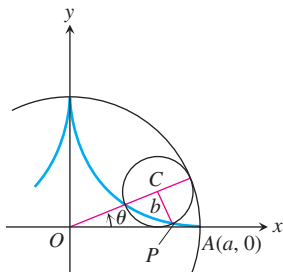
Exercises 1–12 give parametric equations and parameter intervals for the motion of a particle in the xy -plane. Identify the particle's path by finding a Cartesian equation for it. Graph the Cartesian equation. (The graphs will vary with the equation used.) Indicate the portion of the graph traced by the particle and the direction of motion.

1. $x = \cos t$, $y = \sin t$, $0 \leq t \leq \pi$
2. $x = \sin(2\pi(1 - t))$, $y = \cos(2\pi(1 - t))$; $0 \leq t \leq 1$
3. $x = 4 \cos t$, $y = 5 \sin t$; $0 \leq t \leq \pi$
4. $x = 4 \sin t$, $y = 5 \cos t$; $0 \leq t \leq 2\pi$
5. $x = t$, $y = \sqrt{t}$; $t \geq 0$
6. $x = \sec^2 t - 1$, $y = \tan t$; $-\pi/2 < t < \pi/2$

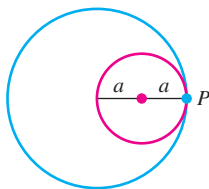
7. $x = -\sec t$, $y = \tan t$; $-\pi/2 < t < \pi/2$
8. $x = \csc t$, $y = \cot t$; $0 < t < \pi$
9. $x = t$, $y = \sqrt{4 - t^2}$; $0 \leq t \leq 2$
10. $x = t^2$, $y = \sqrt{t^4 + 1}$; $t \geq 0$
11. $x = -\cosh t$, $y = \sinh t$; $-\infty < t < \infty$
12. $x = 2 \sinh t$, $y = 2 \cosh t$; $-\infty < t < \infty$
13. **Hypocycloids** When a circle rolls on the inside of a fixed circle, any point P on the circumference of the rolling circle describes a *hypocycloid*. Let the fixed circle be $x^2 + y^2 = a^2$, let the radius of the rolling circle be b , and let the initial position of the tracing point P be $A(a, 0)$. Find parametric equations for the hypocycloid, using as the parameter the angle θ from the positive x -axis to the line joining the circles' centers. In particular, if

$b = a/4$, as in the accompanying figure, show that the hypocycloid is the astroid

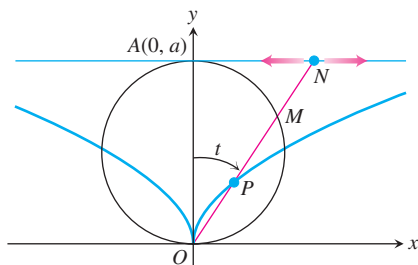
$$x = a \cos^3 \theta, \quad y = a \sin^3 \theta.$$



- 14. More about hypocycloids** The accompanying figure shows a circle of radius a tangent to the inside of a circle of radius $2a$. The point P , shown as the point of tangency in the figure, is attached to the smaller circle. What path does P trace as the smaller circle rolls around the inside of the larger circle?



- 15.** As the point N moves along the line $y = a$ in the accompanying figure, P moves in such a way that $OP = MN$. Find parametric equations for the coordinates of P as functions of the angle t that the line ON makes with the positive y -axis.



- 16. Trochoids** A wheel of radius a rolls along a horizontal straight line without slipping. Find parametric equations for the curve traced out by a point P on a spoke of the wheel b units from its center. As parameter, use the angle θ through which the wheel turns. The curve is called a *trochoid*, which is a cycloid when $b = a$.

Distance Using Parametric Equations

- 17.** Find the point on the parabola $x = t, y = t^2, -\infty < t < \infty$, closest to the point $(2, 1/2)$. (Hint: Minimize the square of the distance as a function of t .)

- 18.** Find the point on the ellipse $x = 2 \cos t, y = \sin t, 0 \leq t \leq 2\pi$ closest to the point $(3/4, 0)$. (Hint: Minimize the square of the distance as a function of t .)

T GRAPHER EXPLORATIONS

If you have a parametric equation grapher, graph the following equations over the given intervals.

- 19. Ellipse** $x = 4 \cos t, y = 2 \sin t$, over
 a. $0 \leq t \leq 2\pi$ b. $0 \leq t \leq \pi$
 c. $-\pi/2 \leq t \leq \pi/2$.
- 20. Hyperbola branch** $x = \sec t$ (enter as $1/\cos(t)$), $y = \tan t$ (enter as $\sin(t)/\cos(t)$), over
 a. $-1.5 \leq t \leq 1.5$ b. $-0.5 \leq t \leq 0.5$
 c. $-0.1 \leq t \leq 0.1$.
- 21. Parabola** $x = 2t + 3, y = t^2 - 1, -2 \leq t \leq 2$
- 22. Cycloid** $x = t - \sin t, y = 1 - \cos t$, over
 a. $0 \leq t \leq 2\pi$ b. $0 \leq t \leq 4\pi$
 c. $\pi \leq t \leq 3\pi$.

23. A nice curve (a deltoid)

$$x = 2 \cos t + \cos 2t, \quad y = 2 \sin t - \sin 2t; \quad 0 \leq t \leq 2\pi$$

What happens if you replace 2 with -2 in the equations for x and y ? Graph the new equations and find out.

24. An even nicer curve

$$x = 3 \cos t + \cos 3t, \quad y = 3 \sin t - \sin 3t; \quad 0 \leq t \leq 2\pi$$

What happens if you replace 3 with -3 in the equations for x and y ? Graph the new equations and find out.

25. Three beautiful curves

a. *Epicycloid*:

$$x = 9 \cos t - \cos 9t, \quad y = 9 \sin t - \sin 9t; \quad 0 \leq t \leq 2\pi$$

b. *Hypocycloid*:

$$x = 8 \cos t + 2 \cos 4t, \quad y = 8 \sin t - 2 \sin 4t; \quad 0 \leq t \leq 2\pi$$

c. *Hypotrochoid*:

$$x = \cos t + 5 \cos 3t, \quad y = 6 \cos t - 5 \sin 3t; \quad 0 \leq t \leq 2\pi$$

26. More beautiful curves

a. $x = 6 \cos t + 5 \cos 3t, \quad y = 6 \sin t - 5 \sin 3t;$
 $0 \leq t \leq 2\pi$

b. $x = 6 \cos 2t + 5 \cos 6t, \quad y = 6 \sin 2t - 5 \sin 6t;$
 $0 \leq t \leq \pi$

c. $x = 6 \cos t + 5 \cos 3t, \quad y = 6 \sin 2t - 5 \sin 3t;$
 $0 \leq t \leq 2\pi$

d. $x = 6 \cos 2t + 5 \cos 6t, \quad y = 6 \sin 4t - 5 \sin 6t;$
 $0 \leq t \leq \pi$

10.5
Polar Coordinates

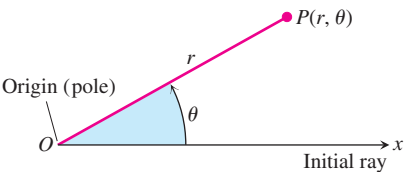


FIGURE 10.35 To define polar coordinates for the plane, we start with an origin, called the pole, and an initial ray.

In this section, we study polar coordinates and their relation to Cartesian coordinates. While a point in the plane has just one pair of Cartesian coordinates, it has infinitely many pairs of polar coordinates. This has interesting consequences for graphing, as we will see in the next section.

Definition of Polar Coordinates

To define polar coordinates, we first fix an **origin** O (called the **pole**) and an **initial ray** from O (Figure 10.35). Then each point P can be located by assigning to it a **polar coordinate pair** (r, θ) in which r gives the directed distance from O to P and θ gives the directed angle from the initial ray to ray OP .

Polar Coordinates

$P(r, \theta)$

Directed distance from O to P

Directed angle from initial ray to OP

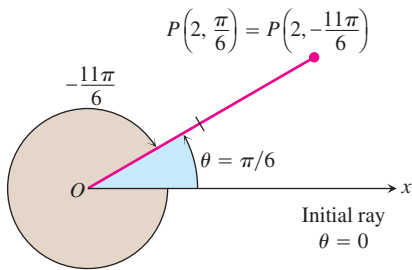


FIGURE 10.36 Polar coordinates are not unique.

As in trigonometry, θ is positive when measured counterclockwise and negative when measured clockwise. The angle associated with a given point is not unique. For instance, the point 2 units from the origin along the ray $\theta = \pi/6$ has polar coordinates $r = 2$, $\theta = \pi/6$. It also has coordinates $r = 2$, $\theta = -11\pi/6$ (Figure 10.36). There are occasions when we wish to allow r to be negative. That is why we use directed distance in defining $P(r, \theta)$. The point $P(2, 7\pi/6)$ can be reached by turning $7\pi/6$ radians counterclockwise from the initial ray and going forward 2 units (Figure 10.37). It can also be reached by turning $\pi/6$ radians counterclockwise from the initial ray and going *backward* 2 units. So the point also has polar coordinates $r = -2$, $\theta = \pi/6$.

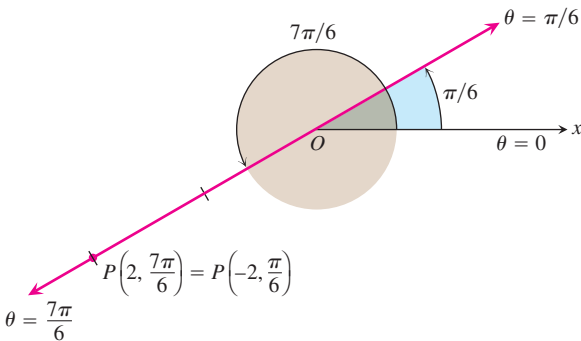


FIGURE 10.37 Polar coordinates can have negative r -values.

EXAMPLE 1 Finding Polar Coordinates

Find all the polar coordinates of the point $P(2, \pi/6)$.

Solution We sketch the initial ray of the coordinate system, draw the ray from the origin that makes an angle of $\pi/6$ radians with the initial ray, and mark the point $(2, \pi/6)$ (Figure 10.38). We then find the angles for the other coordinate pairs of P in which $r = 2$ and $r = -2$.

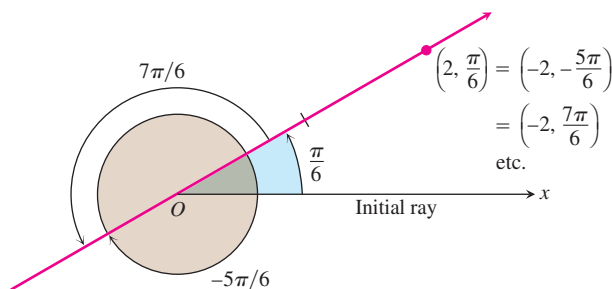


FIGURE 10.38 The point $P(2, \pi/6)$ has infinitely many polar coordinate pairs (Example 1).

For $r = 2$, the complete list of angles is

$$\frac{\pi}{6}, \quad \frac{\pi}{6} \pm 2\pi, \quad \frac{\pi}{6} \pm 4\pi, \quad \frac{\pi}{6} \pm 6\pi, \quad \dots$$

For $r = -2$, the angles are

$$-\frac{5\pi}{6}, \quad -\frac{5\pi}{6} \pm 2\pi, \quad -\frac{5\pi}{6} \pm 4\pi, \quad -\frac{5\pi}{6} \pm 6\pi, \quad \dots$$

The corresponding coordinate pairs of P are

$$\left(2, \frac{\pi}{6} + 2n\pi\right), \quad n = 0, \pm 1, \pm 2, \dots$$

and

$$\left(-2, -\frac{5\pi}{6} + 2n\pi\right), \quad n = 0, \pm 1, \pm 2, \dots$$

When $n = 0$, the formulas give $(2, \pi/6)$ and $(-2, -5\pi/6)$. When $n = 1$, they give $(2, 13\pi/6)$ and $(-2, 7\pi/6)$, and so on. ■

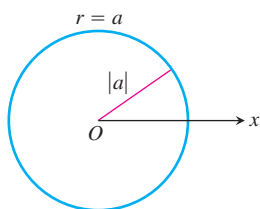


FIGURE 10.39 The polar equation for a circle is $r = a$.

Polar Equations and Graphs

If we hold r fixed at a constant value $r = a \neq 0$, the point $P(r, \theta)$ will lie $|a|$ units from the origin O . As θ varies over any interval of length 2π , P then traces a circle of radius $|a|$ centered at O (Figure 10.39).

If we hold θ fixed at a constant value $\theta = \theta_0$ and let r vary between $-\infty$ and ∞ , the point $P(r, \theta)$ traces the line through O that makes an angle of measure θ_0 with the initial ray.

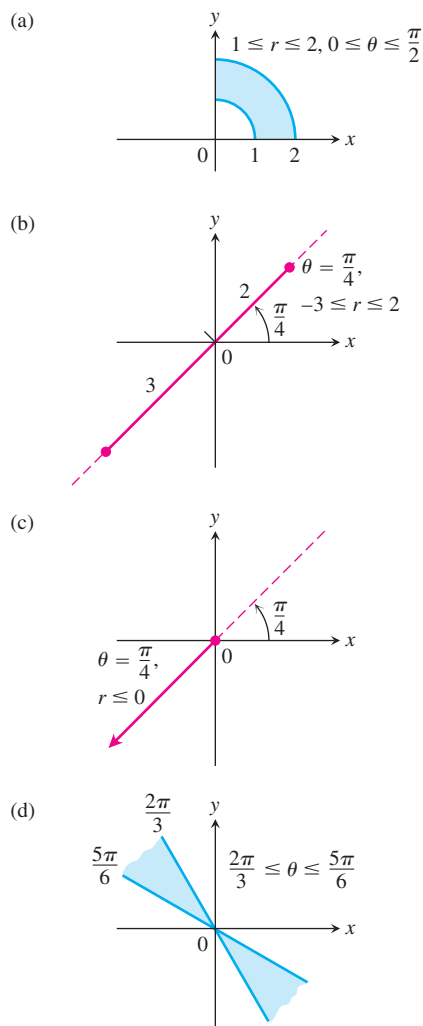


FIGURE 10.40 The graphs of typical inequalities in r and θ (Example 3).

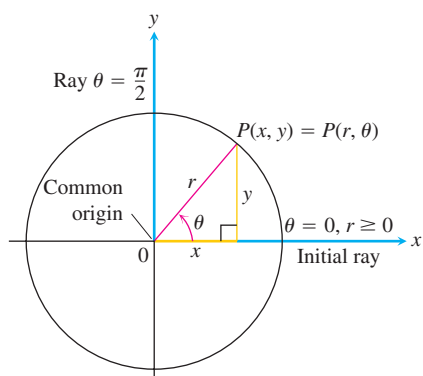


FIGURE 10.41 The usual way to relate polar and Cartesian coordinates.

Equation

$$r = a$$

$$\theta = \theta_0$$

Graph

Circle radius $|a|$ centered at O

Line through O making an angle θ_0 with the initial ray

EXAMPLE 2 Finding Polar Equations for Graphs

(a) $r = 1$ and $r = -1$ are equations for the circle of radius 1 centered at O .

(b) $\theta = \pi/6$, $\theta = 7\pi/6$, and $\theta = -5\pi/6$ are equations for the line in Figure 10.38.

Equations of the form $r = a$ and $\theta = \theta_0$ can be combined to define regions, segments, and rays.

EXAMPLE 3 Identifying Graphs

Graph the sets of points whose polar coordinates satisfy the following conditions.

(a) $1 \leq r \leq 2$ and $0 \leq \theta \leq \frac{\pi}{2}$

(b) $-3 \leq r \leq 2$ and $\theta = \frac{\pi}{4}$

(c) $r \leq 0$ and $\theta = \frac{\pi}{4}$

(d) $\frac{2\pi}{3} \leq \theta \leq \frac{5\pi}{6}$ (no restriction on r)

Solution The graphs are shown in Figure 10.40.

Relating Polar and Cartesian Coordinates

When we use both polar and Cartesian coordinates in a plane, we place the two origins together and take the initial polar ray as the positive x -axis. The ray $\theta = \pi/2$, $r > 0$, becomes the positive y -axis (Figure 10.41). The two coordinate systems are then related by the following equations.

Equations Relating Polar and Cartesian Coordinates

$$x = r \cos \theta, \quad y = r \sin \theta, \quad x^2 + y^2 = r^2$$

The first two of these equations uniquely determine the Cartesian coordinates x and y given the polar coordinates r and θ . On the other hand, if x and y are given, the third equation gives two possible choices for r (a positive and a negative value). For each selection, there is a unique $\theta \in [0, 2\pi)$ satisfying the first two equations, each then giving a polar coordinate representation of the Cartesian point (x, y) . The other polar coordinate representations for the point can be determined from these two, as in Example 1.

EXAMPLE 4 Equivalent Equations

Polar equation	Cartesian equivalent
$r \cos \theta = 2$	$x = 2$
$r^2 \cos \theta \sin \theta = 4$	$xy = 4$
$r^2 \cos^2 \theta - r^2 \sin^2 \theta = 1$	$x^2 - y^2 = 1$
$r = 1 + 2r \cos \theta$	$y^2 - 3x^2 - 4x - 1 = 0$
$r = 1 - \cos \theta$	$x^4 + y^4 + 2x^2y^2 + 2x^3 + 2xy^2 - y^2 = 0$

With some curves, we are better off with polar coordinates; with others, we aren't. ■

EXAMPLE 5 Converting Cartesian to Polar

Find a polar equation for the circle $x^2 + (y - 3)^2 = 9$ (Figure 10.42).

Solution

$$\begin{aligned}
 x^2 + y^2 - 6y + 9 &= 9 && \text{Expand } (y - 3)^2. \\
 x^2 + y^2 - 6y &= 0 && \text{The 9's cancel.} \\
 r^2 - 6r \sin \theta &= 0 && x^2 + y^2 = r^2 \\
 r = 0 \quad \text{or} \quad r - 6 \sin \theta &= 0 \\
 r &= 6 \sin \theta && \text{Includes both possibilities}
 \end{aligned}$$

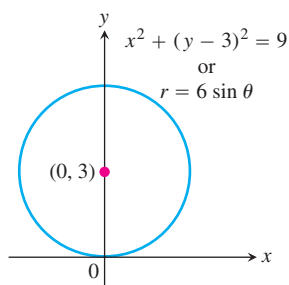


FIGURE 10.42 The circle in Example 5.

We will say more about polar equations of conic sections in Section 10.8. ■

EXAMPLE 6 Converting Polar to Cartesian

Replace the following polar equations by equivalent Cartesian equations, and identify their graphs.

- (a) $r \cos \theta = -4$
 (b) $r^2 = 4r \cos \theta$
 (c) $r = \frac{4}{2 \cos \theta - \sin \theta}$

Solution We use the substitutions $r \cos \theta = x$, $r \sin \theta = y$, $r^2 = x^2 + y^2$.

- (a) $r \cos \theta = -4$

The Cartesian equation: $r \cos \theta = -4$
 $x = -4$

The graph: Vertical line through $x = -4$ on the x -axis

- (b) $r^2 = 4r \cos \theta$

The Cartesian equation: $r^2 = 4r \cos \theta$
 $x^2 + y^2 = 4x$
 $x^2 - 4x + y^2 = 0$
 $x^2 - 4x + 4 + y^2 = 4$ Completing the square
 $(x - 2)^2 + y^2 = 4$

The graph: Circle, radius 2, center $(h, k) = (2, 0)$

$$(c) \quad r = \frac{4}{2 \cos \theta - \sin \theta}$$

The Cartesian equation: $r(2 \cos \theta - \sin \theta) = 4$

$$2r \cos \theta - r \sin \theta = 4$$

$$2x - y = 4$$

$$y = 2x - 4$$

The graph: Line, slope $m = 2$, y -intercept $b = -4$



EXERCISES 10.5

Polar Coordinate Pairs

- Which polar coordinate pairs label the same point?
 - $(3, 0)$
 - $(-3, 0)$
 - $(2, 2\pi/3)$
 - $(2, 7\pi/3)$
 - $(-3, \pi)$
 - $(2, \pi/3)$
 - $(-3, 2\pi)$
 - $(-2, -\pi/3)$
- Which polar coordinate pairs label the same point?
 - $(-2, \pi/3)$
 - $(2, -\pi/3)$
 - (r, θ)
 - $(r, \theta + \pi)$
 - $(-r, \theta)$
 - $(2, -2\pi/3)$
 - $(-r, \theta + \pi)$
 - $(-2, 2\pi/3)$
- Plot the following points (given in polar coordinates). Then find all the polar coordinates of each point.
 - $(2, \pi/2)$
 - $(2, 0)$
 - $(-2, \pi/2)$
 - $(-2, 0)$
- Plot the following points (given in polar coordinates). Then find all the polar coordinates of each point.
 - $(3, \pi/4)$
 - $(-3, \pi/4)$
 - $(3, -\pi/4)$
 - $(-3, -\pi/4)$

Polar to Cartesian Coordinates

- Find the Cartesian coordinates of the points in Exercise 1.
- Find the Cartesian coordinates of the following points (given in polar coordinates).
 - $(\sqrt{2}, \pi/4)$
 - $(1, 0)$
 - $(0, \pi/2)$
 - $(-\sqrt{2}, \pi/4)$
 - $(-3, 5\pi/6)$
 - $(5, \tan^{-1}(4/3))$
 - $(-1, 7\pi)$
 - $(2\sqrt{3}, 2\pi/3)$

Graphing Polar Equations and Inequalities

Graph the sets of points whose polar coordinates satisfy the equations and inequalities in Exercises 7–22.

- $r = 2$
- $0 \leq r \leq 2$
- $r \geq 1$
- $1 \leq r \leq 2$
- $0 \leq \theta \leq \pi/6, r \geq 0$
- $\theta = 2\pi/3, r \leq -2$

- $\theta = \pi/3, -1 \leq r \leq 3$
- $\theta = \pi/2, r \geq 0$
- $0 \leq \theta \leq \pi, r = 1$
- $\pi/4 \leq \theta \leq 3\pi/4, 0 \leq r \leq 1$
- $-\pi/4 \leq \theta \leq \pi/4, -1 \leq r \leq 1$
- $-\pi/2 \leq \theta \leq \pi/2, 1 \leq r \leq 2$
- $0 \leq \theta \leq \pi/2, 1 \leq |r| \leq 2$
- $\theta = 11\pi/4, r \geq -1$
- $\theta = \pi/2, r \leq 0$
- $0 \leq \theta \leq \pi, r = -1$

Polar to Cartesian Equations

Replace the polar equations in Exercises 23–48 by equivalent Cartesian equations. Then describe or identify the graph.

- $r \cos \theta = 2$
- $r \sin \theta = -1$
- $r \sin \theta = 0$
- $r \cos \theta = 0$
- $r = 4 \csc \theta$
- $r = -3 \sec \theta$
- $r \cos \theta + r \sin \theta = 1$
- $r \sin \theta = r \cos \theta$
- $r^2 = 1$
- $r^2 = 4r \sin \theta$
- $r = \frac{5}{\sin \theta - 2 \cos \theta}$
- $r^2 \sin 2\theta = 2$
- $r = \cot \theta \csc \theta$
- $r = 4 \tan \theta \sec \theta$
- $r = \csc \theta e^{r \cos \theta}$
- $r \sin \theta = \ln r + \ln \cos \theta$
- $r^2 + 2r^2 \cos \theta \sin \theta = 1$
- $\cos^2 \theta = \sin^2 \theta$
- $r^2 = -4r \cos \theta$
- $r^2 = -6r \sin \theta$
- $r = 8 \sin \theta$
- $r = 3 \cos \theta$
- $r = 2 \cos \theta + 2 \sin \theta$
- $r = 2 \cos \theta - \sin \theta$
- $r \sin \left(\theta + \frac{\pi}{6} \right) = 2$
- $r \sin \left(\frac{2\pi}{3} - \theta \right) = 5$

Cartesian to Polar Equations

Replace the Cartesian equations in Exercises 49–62 by equivalent polar equations.

- $x = 7$
- $y = 1$
- $x = y$
- $x - y = 3$
- $x^2 + y^2 = 4$
- $x^2 - y^2 = 1$
- $\frac{x^2}{9} + \frac{y^2}{4} = 1$
- $xy = 2$

57. $y^2 = 4x$

58. $x^2 + xy + y^2 = 1$

59. $x^2 + (y - 2)^2 = 4$

60. $(x - 5)^2 + y^2 = 25$

61. $(x - 3)^2 + (y + 1)^2 = 4$

62. $(x + 2)^2 + (y - 5)^2 = 16$

Theory and Examples

63. Find all polar coordinates of the origin.

64. Vertical and horizontal lines

- Show that every vertical line in the xy -plane has a polar equation of the form $r = a \sec \theta$.
- Find the analogous polar equation for horizontal lines in the xy -plane.

10.6

Graphing in Polar Coordinates

This section describes techniques for graphing equations in polar coordinates.

Symmetry

Figure 10.43 illustrates the standard polar coordinate tests for symmetry.

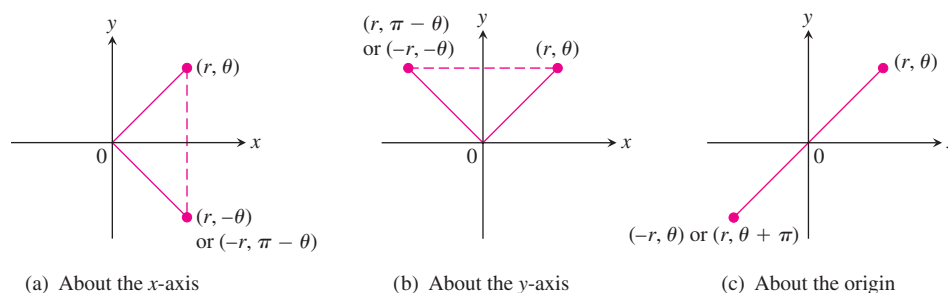


FIGURE 10.43 Three tests for symmetry in polar coordinates.

Symmetry Tests for Polar Graphs

1. *Symmetry about the x -axis:* If the point (r, θ) lies on the graph, the point $(r, -\theta)$ or $(-r, \pi - \theta)$ lies on the graph (Figure 10.43a).
2. *Symmetry about the y -axis:* If the point (r, θ) lies on the graph, the point $(r, \pi - \theta)$ or $(-r, -\theta)$ lies on the graph (Figure 10.43b).
3. *Symmetry about the origin:* If the point (r, θ) lies on the graph, the point $(-r, \theta)$ or $(r, \theta + \pi)$ lies on the graph (Figure 10.43c).

Slope

The slope of a polar curve $r = f(\theta)$ is given by dy/dx , not by $r' = df/d\theta$. To see why, think of the graph of f as the graph of the parametric equations

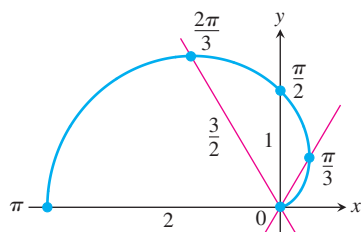
$$x = r \cos \theta = f(\theta) \cos \theta, \quad y = r \sin \theta = f(\theta) \sin \theta.$$

If f is a differentiable function of θ , then so are x and y and, when $dx/d\theta \neq 0$, we can calculate dy/dx from the parametric formula

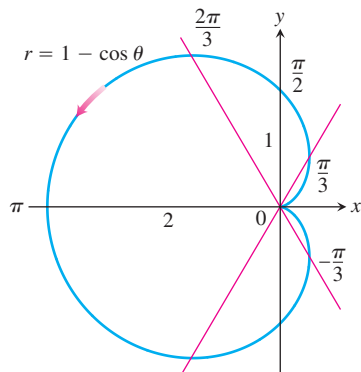
$$\begin{aligned} \frac{dy}{dx} &= \frac{dy/d\theta}{dx/d\theta} && \text{Section 3.5, Equation (2) with } t = \theta \\ &= \frac{\frac{d}{d\theta}(f(\theta) \cdot \sin \theta)}{\frac{d}{d\theta}(f(\theta) \cdot \cos \theta)} \\ &= \frac{\frac{df}{d\theta} \sin \theta + f(\theta) \cos \theta}{\frac{df}{d\theta} \cos \theta - f(\theta) \sin \theta} && \text{Product Rule for derivatives} \end{aligned}$$

θ	$r = 1 - \cos \theta$
0	0
$\frac{\pi}{3}$	$\frac{1}{2}$
$\frac{\pi}{2}$	1
$\frac{2\pi}{3}$	$\frac{3}{2}$
π	2

(a)



(b)



(c)

FIGURE 10.44 The steps in graphing the cardioid $r = 1 - \cos \theta$ (Example 1). The arrow shows the direction of increasing θ .

Slope of the Curve $r = f(\theta)$

$$\left. \frac{dy}{dx} \right|_{(r, \theta)} = \frac{f'(\theta) \sin \theta + f(\theta) \cos \theta}{f'(\theta) \cos \theta - f(\theta) \sin \theta},$$

provided $dx/d\theta \neq 0$ at (r, θ) .

If the curve $r = f(\theta)$ passes through the origin at $\theta = \theta_0$, then $f(\theta_0) = 0$, and the slope equation gives

$$\left. \frac{dy}{dx} \right|_{(0, \theta_0)} = \frac{f'(\theta_0) \sin \theta_0}{f'(\theta_0) \cos \theta_0} = \tan \theta_0.$$

If the graph of $r = f(\theta)$ passes through the origin at the value $\theta = \theta_0$, the slope of the curve there is $\tan \theta_0$. The reason we say “slope at $(0, \theta_0)$ ” and not just “slope at the origin” is that a polar curve may pass through the origin (or any point) more than once, with different slopes at different θ -values. This is not the case in our first example, however.

EXAMPLE 1 A Cardioid

Graph the curve $r = 1 - \cos \theta$.

Solution The curve is symmetric about the x -axis because

$$\begin{aligned} (r, \theta) \text{ on the graph} &\Rightarrow r = 1 - \cos \theta \\ &\Rightarrow r = 1 - \cos(-\theta) && \cos \theta = \cos(-\theta) \\ &\Rightarrow (r, -\theta) \text{ on the graph.} \end{aligned}$$

As θ increases from 0 to π , $\cos \theta$ decreases from 1 to -1 , and $r = 1 - \cos \theta$ increases from a minimum value of 0 to a maximum value of 2. As θ continues on from π to 2π , $\cos \theta$ increases from -1 back to 1 and r decreases from 2 back to 0. The curve starts to repeat when $\theta = 2\pi$ because the cosine has period 2π .

The curve leaves the origin with slope $\tan(0) = 0$ and returns to the origin with slope $\tan(2\pi) = 0$.

We make a table of values from $\theta = 0$ to $\theta = \pi$, plot the points, draw a smooth curve through them with a horizontal tangent at the origin, and reflect the curve across the x -axis to complete the graph (Figure 10.44). The curve is called a *cardioid* because of its heart shape. Cardioid shapes appear in the cams that direct the even layering of thread on bobbins and reels, and in the signal-strength pattern of certain radio antennas. ■

EXAMPLE 2 Graph the Curve $r^2 = 4 \cos \theta$.

Solution The equation $r^2 = 4 \cos \theta$ requires $\cos \theta \geq 0$, so we get the entire graph by running θ from $-\pi/2$ to $\pi/2$. The curve is symmetric about the x -axis because

$$\begin{aligned}(r, \theta) \text{ on the graph} &\Rightarrow r^2 = 4 \cos \theta \\ &\Rightarrow r^2 = 4 \cos(-\theta) && \cos \theta = \cos(-\theta) \\ &\Rightarrow (r, -\theta) \text{ on the graph.}\end{aligned}$$

The curve is also symmetric about the origin because

$$\begin{aligned}(r, \theta) \text{ on the graph} &\Rightarrow r^2 = 4 \cos \theta \\ &\Rightarrow (-r)^2 = 4 \cos \theta \\ &\Rightarrow (-r, \theta) \text{ on the graph.}\end{aligned}$$

Together, these two symmetries imply symmetry about the y -axis.

The curve passes through the origin when $\theta = -\pi/2$ and $\theta = \pi/2$. It has a vertical tangent both times because $\tan \theta$ is infinite.

For each value of θ in the interval between $-\pi/2$ and $\pi/2$, the formula $r^2 = 4 \cos \theta$ gives two values of r :

$$r = \pm 2\sqrt{\cos \theta}.$$

We make a short table of values, plot the corresponding points, and use information about symmetry and tangents to guide us in connecting the points with a smooth curve (Figure 10.45).

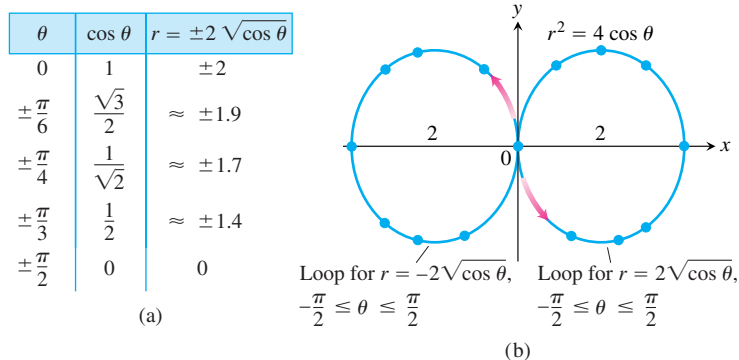


FIGURE 10.45 The graph of $r^2 = 4 \cos \theta$. The arrows show the direction of increasing θ . The values of r in the table are rounded (Example 2).

A Technique for Graphing

One way to graph a polar equation $r = f(\theta)$ is to make a table of (r, θ) -values, plot the corresponding points, and connect them in order of increasing θ . This can work well if enough points have been plotted to reveal all the loops and dimples in the graph. Another method of graphing that is usually quicker and more reliable is to

1. first graph $r = f(\theta)$ in the Cartesian $r\theta$ -plane,
2. then use the Cartesian graph as a “table” and guide to sketch the polar coordinate graph.

This method is better than simple point plotting because the first Cartesian graph, even when hastily drawn, shows at a glance where r is positive, negative, and nonexistent, as well as where r is increasing and decreasing. Here's an example.

EXAMPLE 3 A Lemniscate

Graph the curve

$$r^2 = \sin 2\theta.$$

Solution Here we begin by plotting r^2 (not r) as a function of θ in the Cartesian $r^2\theta$ -plane. See Figure 10.46a. We pass from there to the graph of $r = \pm\sqrt{\sin 2\theta}$ in the $r\theta$ -plane (Figure 10.46b), and then draw the polar graph (Figure 10.46c). The graph in Figure 10.46b “covers” the final polar graph in Figure 10.46c twice. We could have managed with either loop alone, with the two upper halves, or with the two lower halves. The double covering does no harm, however, and we actually learn a little more about the behavior of the function this way. ■

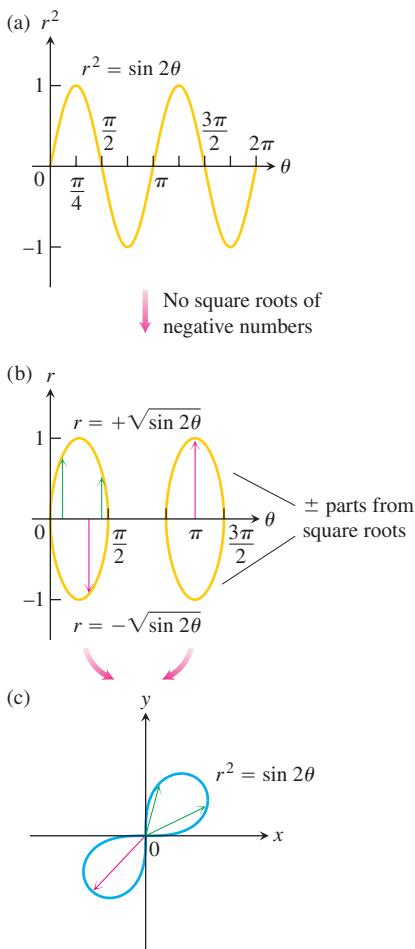


FIGURE 10.46 To plot $r = f(\theta)$ in the Cartesian $r\theta$ -plane in (b), we first plot $r^2 = \sin 2\theta$ in the $r^2\theta$ -plane in (a) and then ignore the values of θ for which $\sin 2\theta$ is negative. The radii from the sketch in (b) cover the polar graph of the lemniscate in (c) twice (Example 3).

Finding Points Where Polar Graphs Intersect

The fact that we can represent a point in different ways in polar coordinates makes extra care necessary in deciding when a point lies on the graph of a polar equation and in determining the points in which polar graphs intersect. The problem is that a point of intersection may satisfy the equation of one curve with polar coordinates that are different from the ones with which it satisfies the equation of another curve. Thus, solving the equations of two curves simultaneously may not identify all their points of intersection. One sure way to identify all the points of intersection is to graph the equations.

EXAMPLE 4 Deceptive Polar Coordinates

Show that the point $(2, \pi/2)$ lies on the curve $r = 2 \cos 2\theta$.

Solution It may seem at first that the point $(2, \pi/2)$ does not lie on the curve because substituting the given coordinates into the equation gives

$$2 = 2 \cos 2\left(\frac{\pi}{2}\right) = 2 \cos \pi = -2,$$

which is not a true equality. The magnitude is right, but the sign is wrong. This suggests looking for a pair of coordinates for the same given point in which r is negative, for example, $(-2, -(\pi/2))$. If we try these in the equation $r = 2 \cos 2\theta$, we find

$$-2 = 2 \cos 2\left(-\frac{\pi}{2}\right) = 2(-1) = -2,$$

and the equation is satisfied. The point $(2, \pi/2)$ does lie on the curve. ■

EXAMPLE 5 Elusive Intersection Points

Find the points of intersection of the curves

$$r^2 = 4 \cos \theta \quad \text{and} \quad r = 1 - \cos \theta.$$

HISTORICAL BIOGRAPHY

Johannes Kepler
(1571–1630)

Solution In Cartesian coordinates, we can always find the points where two curves cross by solving their equations simultaneously. In polar coordinates, the story is different. Simultaneous solution may reveal some intersection points without revealing others. In this example, simultaneous solution reveals only two of the four intersection points. The others are found by graphing. (Also, see Exercise 49.)

If we substitute $\cos \theta = r^2/4$ in the equation $r = 1 - \cos \theta$, we get

$$r = 1 - \cos \theta = 1 - \frac{r^2}{4}$$

$$4r = 4 - r^2$$

$$r^2 + 4r - 4 = 0$$

$$r = -2 \pm 2\sqrt{2}. \quad \text{Quadratic formula}$$

The value $r = -2 - 2\sqrt{2}$ has too large an absolute value to belong to either curve. The values of θ corresponding to $r = -2 + 2\sqrt{2}$ are

$$\begin{aligned} \theta &= \cos^{-1}(1 - r) && \text{From } r = 1 - \cos \theta \\ &= \cos^{-1}(1 - (2\sqrt{2} - 2)) && \text{Set } r = 2\sqrt{2} - 2. \\ &= \cos^{-1}(3 - 2\sqrt{2}) \\ &= \pm 80^\circ. && \text{Rounded to the nearest degree} \end{aligned}$$

We have thus identified two intersection points: $(r, \theta) = (2\sqrt{2} - 2, \pm 80^\circ)$.

If we graph the equations $r^2 = 4 \cos \theta$ and $r = 1 - \cos \theta$ together (Figure 10.47), as we can now do by combining the graphs in Figures 10.44 and 10.45, we see that the curves also intersect at the point $(2, \pi)$ and the origin. Why weren't the r -values of these points revealed by the simultaneous solution? The answer is that the points $(0, 0)$ and $(2, \pi)$ are not on the curves "simultaneously." They are not reached at the same value of θ . On the curve $r = 1 - \cos \theta$, the point $(2, \pi)$ is reached when $\theta = \pi$. On the curve $r^2 = 4 \cos \theta$, it is reached when $\theta = 0$, where it is identified not by the coordinates $(2, \pi)$, which do not satisfy the equation, but by the coordinates $(-2, 0)$, which do. Similarly, the cardioid reaches the origin when $\theta = 0$, but the curve $r^2 = 4 \cos \theta$ reaches the origin when $\theta = \pi/2$. ■

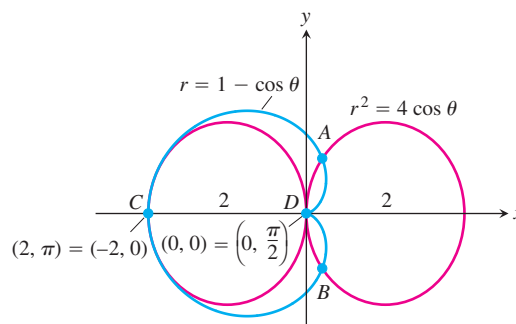


FIGURE 10.47 The four points of intersection of the curves $r = 1 - \cos \theta$ and $r^2 = 4 \cos \theta$ (Example 5). Only A and B were found by simultaneous solution. The other two were disclosed by graphing.

USING TECHNOLOGY Graphing Polar Curves Parametrically

For complicated polar curves we may need to use a graphing calculator or computer to graph the curve. If the device does not plot polar graphs directly, we can convert $r = f(\theta)$ into parametric form using the equations

$$x = r \cos \theta = f(\theta) \cos \theta, \quad y = r \sin \theta = f(\theta) \sin \theta.$$

Then we use the device to draw a parametrized curve in the Cartesian xy -plane. It may be required to use the parameter t rather than θ for the graphing device.